

COMPARATIVE STUDY OF CONVERTER AND COIL TOPOLOGIES FOR AGRICULTURAL DRONE WPT SYSTEMS

Project report submitted to

Visvesvaraya National Institute of Technology, Nagpur

*in partial fulfilment of the requirements for the award of
the degree*

**BACHELOR OF TECHNOLOGY IN
ELECTRICAL AND ELECTRONICS
ENGINEERING**

by

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Saransh Dwivedi (BT22EEE053)

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Om Vaykul (BT22EEE116)

Under the guidance of

Dr. Ritesh Kumar Keshri



**Department of Electrical Engineering
Visvesvaraya National Institute of Technology
Nagpur 440010 (India)
May 2026**

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Declaration

We, Maitreya Takle, Saransh Dwivedi, Sonali Gaikwad, and Om Vaykul hereby declare that this project work titled “**Comparative Study of Converter and Coil Topologies for Agricultural Drone WPT Systems**” is carried out by us in the Department of Electrical Engineering of Visvesvaraya National Institute of Technology, Nagpur. The work is original and has not been submitted earlier whole or in part for the award of any degree/diploma at this or any other Institution / University.

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This is to certify that the project titled “**Comparative Study of Converter and Coil Topologies for Agricultural Drone WPT Systems**”, submitted by **Maitreya Takle, Saransh Dwivedi, Sonali Gaikwad, and Om Vaykul** in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology in Electrical & Electronics Engineering**, VNIT Nagpur. The work is comprehensive, complete and fit for final evaluation.



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Abstract

The rapid expansion of unmanned aerial vehicle deployments in precision agriculture has introduced operational challenges such as frequent battery recharging, where drones must repeatedly interrupt their missions to replenish energy through manual, wired processes. Ensuring continuous and autonomous drone operation is essential for large-scale agricultural productivity and efficiency. When the installed capacity of agricultural drone fleets was limited, the inefficiencies associated with conventional wired charging were considered acceptable due to the minimal disruption to overall operations. The shortcomings of existing charging methods are no longer acceptable given the significant growth in drone adoption and the increasing demand for fully automated, unattended field operations over the past decade. This project proposes a wireless power transfer system designed to enable autonomous, contact-free charging of agricultural drones.

The proposed approach leverages resonant inductive coupling principles and utilises advanced power electronic converter designs to facilitate efficient energy transfer across an air gap between a ground-based charging pad and a drone-mounted receiver. By optimising key parameters such as operating frequency, coil geometry, and compensation network topology, the system achieves reliable power delivery under varying alignment and load conditions. The system is designed to adapt to real-world deployment scenarios including coil misalignment and multi-drone charging configurations. Through extensive simulations and testing using industry-standard tools, the efficiency and reliability of the system are evaluated. The performance metrics include power transfer efficiency, power factor, output voltage regulation, and reduction in charging downtime. The proposed system has also been compared against conventional charging approaches for a more thorough analysis. Overall, this project contributes to advancing the autonomy and operational continuity of agricultural drone systems by providing an efficient and scalable wireless charging solution, thus facilitating the seamless integration of drone technology into modern smart agriculture infrastructure.

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CHAPTER 1

INTRODUCTION

1.1 Unmanned Aerial Vehicles: Types and Applications

Unmanned Aerial Vehicles (UAVs), commonly referred to as drones, have evolved rapidly over the past decade from niche military tools into versatile platforms with widespread civilian and industrial applications. Broadly, drones can be classified by their aerodynamic configuration into three primary types: fixed-wing UAVs, which resemble conventional aircraft and are optimised for long-range endurance missions; rotary-wing UAVs, including multi-rotor platforms such as quadcopters and hexacopters, which are valued for their hovering capability and manoeuvrability in confined spaces; and hybrid VTOL (Vertical Take-Off and Landing) platforms that combine the endurance of fixed-wing designs with the operational flexibility of rotary-wing systems.

Among these, multi-rotor drones have found particular relevance in precision agriculture. Their ability to hover in place, navigate complex terrain, and carry modular payloads makes them well-suited for tasks such as crop health monitoring using multispectral imaging, variable-rate pesticide and fertiliser spraying, field mapping and 3D terrain reconstruction, yield estimation, and real-time data collection for farm management systems. As the global agricultural sector confronts the dual challenge of feeding a growing population while reducing input costs and environmental impact, the adoption of agricultural drones has grown significantly, offering measurable gains in operational efficiency and resource utilisation.

The energy source powering these drones is almost universally the Lithium Polymer (LiPo) or Lithium-Ion battery pack. Charging of these batteries is presently accomplished through one of three methods: direct wired charging via a DC power supply connected to the battery management system; battery hot-swapping, where a discharged pack is physically replaced by a pre-charged one at a ground station; or, in a limited number of commercial deployments, solar-assisted charging integrated into the ground station. Each of these approaches carries inherent limitations that constrain the operational continuity of drone-based agricultural systems.

1.2 Limitations of Conventional Charging Methods

Despite the operational promise of agricultural drones, their practical deployment at scale is severely limited by the constraints of current charging paradigms. A typical agricultural multi-rotor drone, such as the hexacopter considered in this project, carries a

battery pack rated at approximately 21,000 mAh at 22.2 V and can sustain meaningful flight for only 15 to 20 minutes under a working payload. This necessitates frequent returns to a charging station, during which the drone is entirely out of operation.

Wired charging introduces the most significant operational friction. It requires a human operator to physically handle the battery, connect charging cables, and wait through a charging cycle that may span 45 minutes to several hours depending on the charge rate and battery capacity. In a large-area agricultural deployment involving multiple drones operating in relay, this process introduces coordinated downtime that reduces the effective coverage rate and increases the labour intensity of operations — partly negating the automation benefits that motivated drone adoption in the first place.

Battery swapping addresses the downtime issue partially by allowing rapid substitution of depleted packs, but it demands a large inventory of battery packs, increases capital expenditure, and still requires human intervention at the exchange point. Furthermore, in remote agricultural locations where power infrastructure may be limited, maintaining a stock of fully charged batteries presents a logistical challenge.

Beyond operational inefficiency, wired connectors are inherently vulnerable to the harsh environmental conditions of agricultural settings. Dust accumulation, moisture ingress, mud contamination, and mechanical wear on connector pins lead to degraded contact quality, elevated contact resistance, and, in worst cases, arcing or connection failure. This not only increases maintenance costs but introduces safety concerns that are unacceptable in unattended autonomous deployments.

The cumulative effect of these limitations is a system that, despite its technological sophistication, remains bound to periodic human intervention and experiences significant effective downtime — a fundamental bottleneck that must be addressed if agricultural drone systems are to realise their full autonomous potential.

1.3 Wireless Charging as the Solution

Wireless Power Transfer (WPT) technology offers an elegant and robust solution to the charging challenges outlined above. By transferring electrical energy through time-varying magnetic fields rather than physical metallic contacts, WPT eliminates the connector interface entirely. In a drone-based deployment, this means that a drone can autonomously land on a designated charging pad installed in the field or at a base station, and charging commences immediately — without any human intervention, cable connection, or battery removal.

The physical principle underlying WPT is electromagnetic induction, formalised through Faraday's law. A transmitting coil on the charging pad, driven by a high-frequency alternating current, generates a time-varying magnetic flux. When a receiving coil mounted on the drone is brought into proximity with this field, the varying flux induces an electromotive force (EMF) in the receiver, which is subsequently rectified and regulated to charge the onboard battery. By operating both coils at or near their resonant frequency

— in this project, 85 kHz in alignment with the SAE J2954 standard for wireless EV charging — the system achieves resonant inductive coupling, which dramatically improves power transfer efficiency and extends the effective charging distance compared to non-resonant designs.

1.4 Advantages of Wireless Charging for Agricultural Drones

The selection of wireless charging as the energy replenishment strategy for the proposed agricultural drone system is motivated by a convergence of technical and operational advantages that are particularly well-matched to the demands of field deployment.

The most significant advantage is full automation of the recharging process. Once a wireless charging pad is installed at a field station, the drone can execute autonomous landing, initiate charging, and resume its mission without any human involvement. This enables continuous, unattended operation across extended time periods — a prerequisite for large-scale precision agriculture applications.

The absence of exposed electrical contacts directly addresses the environmental durability problem. Agricultural environments expose all equipment to dust, rain, humidity, and biological matter. A connector-free interface has no surface susceptible to corrosion, contamination, or mechanical wear, resulting in substantially lower maintenance requirements and a longer operational service life.

Wireless charging also enables a high degree of design flexibility. The receiving coils can be distributed across the drone's undercarriage — as explored in this project through a 4:1 multi-coil configuration — improving tolerance to landing misalignment without requiring precision docking mechanisms. This is particularly relevant in agricultural settings where the landing surface may be uneven and GPS-guided landing may carry positional uncertainty of several centimetres.

From a system efficiency standpoint, the adoption of a Totem-Pole Power Factor Correction (PFC) front-end — as developed in this work — ensures that the AC-DC conversion stage at the charger maintains near-unity power factor and high conversion efficiency, reducing grid-side harmonic distortion and improving overall energy utilisation. Simulation results from the Totem-Pole PFC stage confirm an efficiency of approximately 98.9% and a power factor approaching unity, underscoring the viability of this architecture for high-power drone charging applications.

In summary, wireless charging transforms the agricultural drone from a system that requires periodic attended maintenance into a genuinely autonomous platform. By removing the manual intervention point from the energy replenishment cycle, it enables drones to contribute continuously to crop monitoring, spraying, and data collection — directly supporting the vision of smart, sustainable, and highly productive modern agriculture.

CHAPTER 2

WIRELESS POWER TRANSFER

2.1 Introduction to Wireless Power Transfer

Wireless Power Transfer (WPT) is a technology that enables the transmission of electrical energy without physical conductors. Instead of wires, it uses electromagnetic fields to transfer energy between a power source and a load across an air gap. First explored by Nikola Tesla, WPT has gained attention due to advancements in power electronics and resonant technologies.

In modern systems, WPT is mainly based on near-field techniques such as inductive and resonant inductive coupling. An alternating current in the transmitting coil generates a magnetic field, which induces voltage in a nearby receiving coil. This energy is then converted into usable electrical power. Compared to wired systems, WPT offers improved safety, reduced wear, and greater convenience. It also supports compact system design and enables fully automated charging without human intervention.

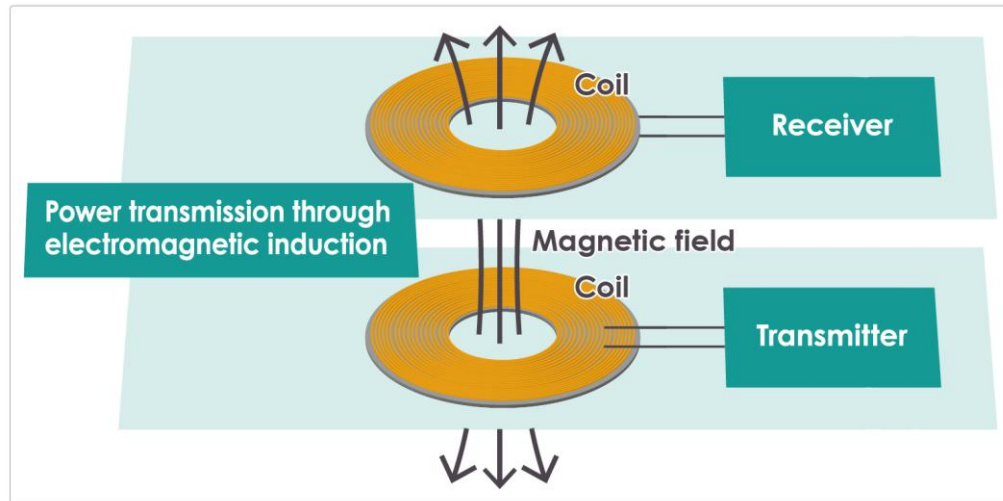


Fig.2.1: Basic Concept of Wireless Power Transfer

The increasing demand for portable electronics, electric vehicles, and autonomous systems has accelerated WPT development. It is especially useful in environments where wired connections are impractical, such as agricultural fields, industrial setups, and sealed electronic systems, making it a key technology for future energy systems.

2.2 Classification of Wireless Power Transfer Techniques

Wireless Power Transfer (WPT) techniques are broadly classified based on the distance of transmission and the physical mechanism used for energy transfer. According to several review papers and surveys, WPT can be divided into two main categories: near-field (non-radiative) and far-field (radiative) techniques.

2.2.1 Near-Field Techniques

Near-field methods operate over short distances (typically less than a few times the coil diameter) and are based on electromagnetic coupling without radiation. These techniques are widely used due to their high efficiency and safety.

a) Inductive Coupling

Inductive coupling is the most fundamental WPT method, where power is transferred through a magnetic field between two coils. An alternating current in the transmitter coil produces a time-varying magnetic field, which induces a voltage in the receiver coil based on Faraday's law.

- ❖ Very high efficiency at short distances
- ❖ Commonly used in chargers for smartphones, toothbrushes, and low-power devices
- ❖ Requires tight alignment between coils

b) Resonant Inductive Coupling (Magnetic Resonance)

This is an advanced form of inductive coupling where both transmitter and receiver are tuned to the same resonant frequency. Resonance significantly improves coupling and allows power transfer over larger distances compared to simple inductive systems.

- ❖ Higher efficiency over mid-range distances
- ❖ Used in electric vehicles, drones, and biomedical implants
- ❖ Allows some tolerance to misalignment

c) Capacitive Coupling

Capacitive WPT uses electric fields between conductive plates instead of magnetic fields. Energy is transferred through displacement current between plates.

- ❖ Suitable for low-power applications
- ❖ Less commonly used than inductive methods
- ❖ Requires high operating frequencies

2.2.2 Far-Field Techniques

Far-field methods are used for long-distance power transfer and involve the transmission of energy in the form of electromagnetic waves such as radio waves, microwaves, or light.

a) Microwave Power Transfer

In this technique, electrical energy is converted into microwaves and transmitted through space using antennas. At the receiver, a rectifying antenna (rectenna) converts the microwave energy back into DC power.

- ❖ Suitable for long-distance applications
- ❖ Lower efficiency compared to near-field methods
- ❖ Used in space-based solar power and experimental systems

b) Laser (Optical) Power Transfer

Electrical energy is converted into a laser beam and transmitted to a photovoltaic receiver, which converts it back into electrical energy.

- ❖ Highly directional and long-range
- ❖ Requires precise alignment
- ❖ Safety concerns due to high-intensity beams

2.3 Working Principle of Wireless Power Transfer

Wireless Power Transfer (WPT) operates on the principle of electromagnetic induction, where energy is transferred through a time-varying magnetic field between two coils without physical contact. The working can be understood through the following key stages:

2.3.1 Generation and Transmission of Magnetic Field

At the transmitter side, an alternating current (AC) is supplied to the transmitting coil, which generates a time-varying magnetic field around it. This magnetic field acts as the medium for energy transfer and propagates through the surrounding space. The strength and stability of this field depend on factors such as input current, coil geometry, number of turns, and operating frequency. High-frequency AC is often used to improve efficiency, reduce the size of passive components, and enhance power density. In practical systems, a high-frequency inverter is employed to convert DC input into the required AC excitation for the transmitting coil.

2.3.2 Magnetic Coupling and Energy Transfer

When a receiving coil is placed within the magnetic field, the changing magnetic flux links the transmitter and receiver coils, creating mutual inductance. According to Faraday's law, this varying flux induces an alternating voltage in the receiving coil, resulting in current flow and energy transfer. The effectiveness of this process depends on the coupling coefficient, which is influenced by the distance, alignment, orientation, and physical dimensions of the coils. To enhance performance, many systems use resonant inductive coupling, where both coils are tuned to the same resonant frequency using compensation networks. This resonance condition minimizes reactive losses, maximizes power transfer, and allows efficient operation even under slight misalignment or varying load conditions.

2.3.3 Power Conversion and Utilization

The induced voltage at the receiver is typically in AC form and must be converted into usable DC. This is achieved using rectifier circuits, followed by filtering to smooth the output and voltage regulation to match the requirements of the load. Additional DC-DC converters may be used to step up or step down the voltage depending on application needs. The converted DC power is then used to charge batteries or directly power electronic devices. Efficient power conditioning at this stage is crucial to ensure stable output, minimize losses, and maintain overall system performance.

2.4 Key Parameters Affecting WPT

The performance of a Wireless Power Transfer (WPT) system is influenced by several electrical and electromagnetic parameters. These parameters determine the efficiency, reliability, and effectiveness of power transfer and must be carefully optimized during system design.

2.4.1 Coupling Coefficient (k)

The coupling coefficient k represents the degree of magnetic coupling between the transmitter (Tx) and receiver (Rx) coils and is defined as:

$$k = \frac{M}{\sqrt{L_1 L_2}} \quad \text{----- 2.4.1}$$

where M is the mutual inductance, and L_1 , L_2 are the self-inductances of the transmitter and receiver coils. A higher value of k indicates stronger magnetic linkage and better energy transfer. In practical systems, k is always less than 1 due to air gaps, leakage flux, and misalignment.

2.4.2 Distance and Alignment

The separation distance between the Tx and Rx coils significantly affects mutual inductance and overall system efficiency. As the distance increases, the magnetic field strength decreases rapidly, resulting in lower induced voltage at the receiver. Additionally, lateral or angular misalignment reduces flux linkage, further decreasing the coupling coefficient. Therefore, WPT systems are typically designed to operate within a limited range where effective coupling can be maintained.

2.4.3 Operating Frequency (f)

The operating frequency plays a crucial role in determining the behaviour of the WPT system. Most systems operate at resonance to maximize power transfer efficiency. The resonant frequency is given by:

$$f = \frac{1}{2\pi\sqrt{LC}} \quad \text{-----2.4.3}$$

where L is the inductance and C is the compensation capacitance.

In this project, the system operates at an **85 kHz** frequency, which lies in the widely accepted range for wireless power transfer applications. This frequency provides an optimal trade-off between efficiency, component size, and switching losses. Operating at 85 kHz enables compact coil design, improved power density, and reduced material usage while maintaining manageable thermal and electromagnetic interference (EMI) effects.

2.4.4 System Efficiency (η)

The efficiency of a WPT system is defined as the ratio of output power at the receiver to input power at the transmitter:

$$\eta = \frac{P_{out}}{P_{in}} \times 100 \% \quad \text{-----2.4.4}$$

Efficiency is affected by various losses, including copper losses in coils, switching losses in power electronics, and leakage flux. High-quality factor (Q) coils, proper impedance matching, and operation at resonance are essential to achieving high efficiency.

2.5 Advantages and Limitations of Wireless Power Transfer

2.5.1 Reduced Downtime and Automation:

Enables automatic charging when the drone lands on the pad, minimizing manual intervention and improving operational efficiency. This allows continuous field operations with minimal human involvement, increasing productivity in large-scale agricultural applications.

2.5.2 No Physical Connectors:

Eliminates wear, corrosion, and contact failures, making the system more reliable in outdoor agricultural environments. It also prevents issues caused by repeated plugging and unplugging, which can degrade connectors over time.

2.5.3 Improved Safety in Harsh Conditions:

Absence of exposed contacts reduces risks of short circuits, sparks, and damage in dusty, wet, or muddy conditions. This makes the system safer for both equipment and operators, especially in unpredictable environmental conditions.

2.5.4 Tolerance to Misalignment:

Multi-coil configurations allow effective power transfer even if the drone is not perfectly aligned with the charging pad. This reduces the need for precise landing mechanisms and simplifies system design.

2.5.5 Low Maintenance and High Durability:

Fewer mechanical and electrical contact points lead to reduced maintenance requirements and longer system lifespan. As a result, operational costs decrease and system reliability improves over long-term usage.

CHAPTER 3

ARCHITECTURE OF THE WIRELESS POWER TRANSFER SYSTEM

3.1 Overview of the Wireless Power Transfer Architecture

Agricultural drones are a transformative technology in modern farming, it enables spraying of fertilizers with high precision, large-area field mapping with minimal human intervention and monitors health of the crops in real time. However, limited energy capacity of the on-board batteries is a challenge for the operational utility of the drone. Most of the agricultural drones get discharged within 15-20 minutes, and it is not reliable to use conventional charger in agricultural environment conditions like dust, moisture, and mud because of the risk of connector corrosion and contamination. Wireless Power Transfer (WPT) offers a compelling solution by enabling autonomous, contact-free charging: the drone just have to lands on a designated charging pad, and energy is transferred electromagnetically from the ground station to the drone without any physical electrical connection.

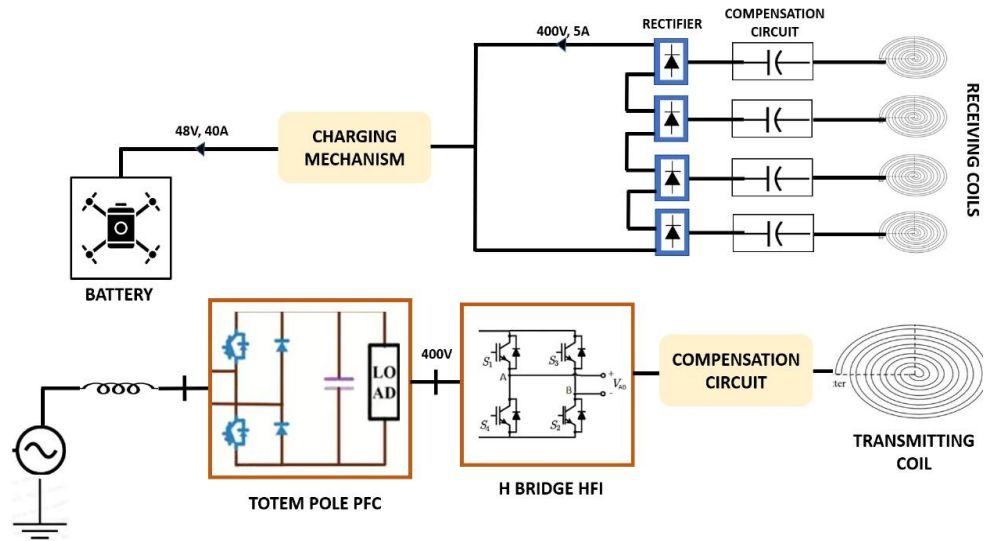


Fig.3.1: Block diagram of architecture of a WPT system

The block diagram in **figure** shows the complete wireless charging system designed for a drone. The system begins with an AC input supply, which is first processed by a Totem-Pole Power Factor Correction (PFC), which regulates and conditions the AC supply into a regulated 400 V DC bus and also maintains the power factor near unity. This DC voltage is then supplied to the High-Frequency Inverter (H Bridge HFI), which

converts the 400 V DC into a high-frequency AC signal required for wireless power transfer. The high-frequency AC is then fed to the transmitter-side compensation circuit, which helps in tuning the system for maximum power transfer and energizes the transmitting coil. When the drone lands on the charging pad, its receiving coils pick up the magnetic field and convert it back into AC power. This received AC is passed through the receiver-side compensation circuit and then rectified into stable DC voltage. The output DC (400 V, 5 A) is fed into the charging mechanism, that steps the rectified 400 V, 5 A output down to the required 48 V, 40 A needed to safely charge the drone's battery.

Table 3.1: System Parameters of the Wireless Drone Charging System

Parameter	Value
AC Input Supply	Single-phase, 230 V AC, 50 Hz
PFC Output (DC Bus Voltage)	400 V DC (regulated)
PFC Topology	Totem-Pole (Bridgeless)
HFI Topology	Full H-Bridge (4 switch)
Operating Frequency	85 kHz
Coil Configuration	1 Transmitter: 4 Receivers (series)
Coil Type	Concentric Circular Spiral
Air Gap (d)	3.5 cm
Mean Coil Diameter (D)	10 cm
Transmitter Coil Power Rating	~ 2 kW
Receiver Output (per path)	400 V, 5 A (DC after rectification)
Battery Charging Output	48 V, 40 A (1.92 kW)
Drone Battery	2 × 21,000 mAh, 22.2 V (6S), in series

3.2 Totem-Pole Power Factor Correction (PFC) Stage

3.2.1 Significance and Role in the System

The Power Factor Correction stage is the first functional block of the wireless charging architecture. It acts as a link connecting the AC utility grid to the high-frequency power electronics system. Primary role of PFC is: first, to rectify and regulate the AC mains supply into a stable, high-voltage DC bus of 400 V; and second, to shape the input current waveform such that it is sinusoidal and in phase with the supply voltage, thereby achieving near-unity power factor. In agricultural fields, where motors, irrigation pumps, and instrumentation, may share a common supply point with the charging station, it is important to maintain high power quality at the grid interface.

Without using an active PFC stage, a standard diode rectifier does not draw a smooth current from the AC supply. Instead, it pulls current in short pulses, resulting in a non-sinusoidal waveform. This type of current contains strong harmonic components, mainly odd harmonics such as 150 Hz, 250 Hz, and 350 Hz. These harmonics lead to higher power losses in the distribution system, create voltage distortion, and may disturb other agricultural equipment operating on the same line, which makes the use of a PFC stage essential in high-power (kilowatt-level) chargers

3.2.2 Operating Principle

In this work, a Totem-Pole PFC is selected due to its reduced conduction losses compared to diode-bridge rectifiers. The Totem-Pole PFC topology is built using two sets of switching devices, each operating at different frequencies. The high-frequency switches (S3 and S4) are responsible for carrying out the boost conversion and typically operate in the range of tens to hundreds of kilohertz. In contrast, the low-frequency switches (S1 and S2) function as synchronous rectifiers and switch only at the AC supply frequency (50 Hz), changing state at each zero crossing of the input voltage. During the positive half cycle of the AC input, switch S1 remains continuously turned ON, providing a conduction path. At the same time, S3 operates in PWM mode. When S3 is ON, the boost inductor stores energy from the AC source through the path formed by S1 and S3. When S3 turns OFF, the stored energy in the inductor is transferred to the DC output (around 400 V) through S1 and the conducting switch S4. In the negative half cycle, the operation is mirrored: S2 takes over the role of the low-frequency conduction switch, while S4 performs high-frequency switching, and S3 provides the freewheeling path for inductor current

To regulate the system, a dual closed-loop control scheme is implemented. The outer voltage loop senses the DC output voltage and, using a PI controller, generates a reference signal representing the desired current amplitude. This signal is then shaped into a sinusoidal reference by multiplying it with a normalized form of the input voltage. The inner current loop compares this reference with the actual inductor current and adjusts the

PWM duty cycle accordingly. This control approach allows the converter to maintain a stable output voltage while ensuring that the input current remains sinusoidal and in phase with the supply voltage.

3.3 High-Frequency Inverter (H-Bridge HFI)

3.3.1 Role in the Wireless Charging System

The High-Frequency Inverter (HFI) forms the second key stage in the wireless power transfer system. In this work, the HFI is designed to ensure efficient conversion with minimal switching losses at high frequency. It takes the regulated 400 V DC output obtained from the Totem-Pole PFC stage and converts it into a high-frequency AC signal, typically around 85 kHz, which matches the operating frequency of the WPT setup. This AC output is then supplied to the transmitter-side compensation network, which in turn energizes the transmitting coil to produce an alternating magnetic field required for wireless energy transfer.

The performance of the overall wireless charging system largely depends on the quality of the inverter output waveform. The need for high-frequency AC arises from the fundamental principle of inductive power transfer. When a coil is supplied with DC, it generates only a constant magnetic field, which is incapable of inducing voltage in a secondary coil. In contrast, an alternating current produces a continuously changing magnetic flux. According to Faraday's law of electromagnetic induction, this varying flux induces an electromotive force in the receiver coil. By operating at a high frequency such as 85 kHz, efficient coupling and power transfer between coils can be achieved. Therefore, the inverter must generate a stable and well-controlled high-frequency AC waveform with minimal losses.

3.2 H-Bridge Topology and Operating Principle

For this system, a full-bridge (H-bridge) inverter configuration is employed. The topology uses four switches (S1, S2, S3, and S4) arranged in two legs, with the load—comprising the coil and compensation network—connected between the midpoints of these legs. The switches are driven in complementary diagonal pairs, namely (S1, S4) and (S2, S3), to achieve bidirectional current flow through the load.

During one switching interval, S1 and S4 are turned ON while S2 and S3 remain OFF, allowing current to pass from the positive DC rail through the load in one direction and return to the negative rail. In the next interval, the switching states are reversed: S2 and S3 conduct while S1 and S4 are OFF, causing the current to flow in the opposite direction. By continuously alternating these diagonal pairs at a high switching frequency of 85 kHz, the inverter generates a square-wave AC voltage across the load with a magnitude equal to the DC bus voltage (approximately 400 V). When this voltage is applied to the resonant network formed by the compensation capacitor and transmitter coil

inductance, the circuit suppresses higher-order harmonics and results in a near-sinusoidal current waveform through the coil.

3.4 Transmitting and Receiving Coil Assembly

3.4.1 Role and Significance in the WPT System

The magnetically coupled coil system forms the core of the wireless power transfer setup, as it provides the physical path for energy transfer across the air gap between the ground charging pad and the drone, without any direct electrical connection. The primary (transmitting) coil, located within the charging pad, produces a time-varying magnetic field when excited by the 85 kHz AC output from the H-bridge inverter. The secondary (receiving) coils, positioned on the underside of the drone, capture this varying magnetic field and convert it back into electrical energy through electromagnetic induction, as explained by Faraday's law of electromagnetic induction.

In this work, the coil design is optimized to achieve a balance between efficiency and practical deployment constraints. The overall efficiency of wireless charging is highly dependent on the design of this coil arrangement, often more than any other component in the system. Key parameters such as coil shape, number of turns, type of conductor, operating frequency, air gap distance, and alignment between the coils influence the coupling coefficient and quality factor, which directly affect power transfer capability. The coil assembly is therefore designed not only for electrical performance but also for mechanical compatibility with the drone system. In the case of agricultural drone applications, additional constraints must also be considered, including variations in landing position, size limitations of the charging platform, and weight restrictions for the receiver coils mounted on the drone structure. These practical factors play a significant role in determining the final coil design.

3.5 Compensation Circuit

3.5.1 Need for Compensation in the WPT System

The compensation network consists mainly of capacitive elements placed on both the transmitter and receiver sides of the wireless power transfer system. Its primary role is to achieve resonance with the effective inductance of the coils at the operating frequency of 85 kHz. By doing so, it neutralizes the reactive impedance that would otherwise restrict power flow and reduce the efficiency of the loosely coupled system.

In this application, the coils exhibit weak magnetic coupling because they are separated by an air gap and do not share a magnetic core, unlike conventional transformers where the coupling coefficient is close to unity. As a result, a significant portion of the magnetic flux does not link both coils, leading to high leakage inductance. If no compensation is provided, this inductive effect dominates the circuit behavior, forcing the

inverter to deliver a large reactive current component. This current does not contribute to useful power transfer but increases conduction losses in the switching devices and windings. By introducing properly selected capacitors, the inductive reactance is counteracted at the operating frequency, creating a resonant condition that enables efficient transfer of real power across the system.

3.6 System Integration and End-to-End Power Flow

The charging sequence starts once the drone completes its operation and autonomously lands on the designated charging pad. The system then identifies the presence of the drone, either by sensing a variation in the reflected impedance of the primary circuit or through a communication interface, and initiates the power transfer process. The Totem-Pole PFC stage draws a near-sinusoidal current from the 230 V, 50 Hz supply, achieving a power factor close to unity, and provides a regulated 400 V DC output to the high-frequency inverter with high efficiency.

This DC voltage is then converted into a high-frequency (85 kHz) AC signal by the H-bridge inverter and supplied to the transmitting side resonant circuit. The series compensation capacitor on the transmitter side is tuned to resonate with the transmitting coil at the operating frequency, effectively neutralizing the reactive component of the coil impedance. As a result, a large and nearly sinusoidal current flows through the transmitting coil, producing a strong alternating magnetic field.

When the drone is positioned such that its receiving coils are aligned above the transmitting coil with a small air gap, the alternating magnetic field links with the receiver coils. Based on Faraday's law of electromagnetic induction, this varying magnetic flux induces an AC voltage in each receiving coil. Each receiver circuit is also tuned using a series capacitor to achieve resonance at the same frequency, ensuring maximum power transfer. The induced voltages from the multiple receiver coils are combined to produce a high-voltage AC output, which is then rectified to obtain a DC output of approximately 400 V at a few amperes, corresponding to kilowatt-level power transfer. The rectified DC output is then processed by a DC-DC buck converter, which steps down the voltage to a suitable level required for battery charging. In this system, the converter regulates the output to match the charging requirements of the drone battery pack. A battery management system supervises the charging process by monitoring parameters such as state of charge and ensures safe operation. It also communicates with the ground controller to enable controlled and adaptive charging throughout the process.

CHAPTER 4

INTRODUCTION TO POWER FACTOR AND POWER FACTOR CORRECTION

4.1 What Is Power Factor?

Power factor quantifies how effectively the electrical power supplied by an AC source is actually utilized by a system. It is formally defined as the ratio of Real Power to Apparent Power:

$$\text{Power factor} = \frac{\text{Real Power (kW)}}{\text{Apparent Power (kVA)}} \quad \text{-----4.1.1}$$

In an ideal electrical system, the voltage and current waveforms are perfectly in phase, yielding a power factor of unity. In practice, however, power electronic systems such as rectifiers, converters, and wireless charging circuits inevitably distort and phase-shift the current waveform as a result of continuous diode and MOSFET switching as well as the influence of reactive components. This degradation in waveform quality causes a corresponding reduction in power factor.

A low power factor indicates that the system draws more current than is strictly necessary to deliver the same amount of useful power. The practical consequences of this condition include:

- ❖ Increased losses in the power system
- ❖ Higher heating in cables and components
- ❖ Reduced efficiency of the converter
- ❖ Poor utilization of electrical infrastructure
- ❖ Increased harmonic distortion in the supply

For these reasons, maintaining a high-power factor is a fundamental requirement in modern power electronic system design.

4.2 What Is a Power Factor Correction (PFC) Circuit?

A Power Factor Correction (PFC) circuit is a dedicated power electronic converter whose primary function is to improve the input power factor of a system. This is accomplished by actively shaping the input current waveform so that it closely tracks the profile of the input voltage waveform.

In the majority of practical systems, the AC supply is first converted to DC by means of a rectifier. A standard diode rectifier, however, draws heavily distorted current from the mains, introducing significant harmonic content. To address this, the PFC stage is interposed between the rectifier and the load, where it actively regulates the current drawn from the supply and reshapes it into a sinusoidal waveform.

4.3 Importance of PFC Circuits in Wireless Charging Systems

Wireless charging systems for drones rely on high-frequency power electronic converters to transfer energy through magnetic coupling between a transmitter coil and a receiver coil. These systems inherently involve multiple power conversion stages, including:

- ❖ AC to DC conversion
- ❖ DC voltage regulation
- ❖ High-frequency inverter operation
- ❖ Resonant power transfer

The presence of these cascaded conversion stages means that, without appropriate correction measures, the current drawn from the supply can become severely distorted. This makes the inclusion of a PFC stage not merely beneficial but essential. As a result, the PFC converter serves as the critical first building block of any wireless charging system. Its integration into wireless drone charging platforms offers several distinct advantages:

4.3.1 Improved Efficiency

Wireless charging systems inherently incur losses at the air-gap coupling interface and within the resonant power transfer network. An efficient PFC stage prevents these losses from being compounded further by minimizing additional power dissipation on the input side.

4.3.2 Better Power Quality

PFC circuits actively suppress harmonic content and restore current waveform quality, resulting in a cleaner and more compliant interaction with the utility supply network.

4.3.3 Reduced Stress on Components

Through precise current flow management, PFC circuits help to mitigate overheating and reduce the electrical stress imposed on switches, capacitors, and inductors throughout the converter.

4.3.4 Compliance with Standards

Industrial and commercial power electronic systems must conform to established harmonic emission and power quality standards. The incorporation of PFC circuits is a well-established approach to achieving and maintaining compliance with these regulations.

4.3.5 Stable DC Bus Voltage

The downstream wireless charging inverter demands a well-regulated DC input voltage to operate correctly. The PFC stage performs this function by actively regulating and sustaining the DC bus voltage at the required level.

In summary, the PFC stage serves as an indispensable front-end converter in wireless charging architectures, contributing substantially to the overall efficiency, power quality, and long-term reliability of the charging system.

4.4 Study of Single-phase and Three-phase PFC circuits

In the course of developing the wireless charging system, various Power Factor Correction circuit topologies were examined to gain insight into their operating principles and suitability for high-power applications. The two principal categories of PFC converters investigated in this work are:

- ❖ Single-Phase PFC
- ❖ Three-Phase PFC

The target specification for the wireless drone charging system demands high power handling capability and superior conversion efficiency, both of which favour the adoption of a Three-Phase PFC converter. Nonetheless, approaching the design of a three-phase PFC system directly presents considerable challenges, owing to:

- ❖ Complex control algorithms and increased number of switching devices
- ❖ Three-phase current balancing requirements
- ❖ Advanced PWM generation techniques
- ❖ More complicated sensing and feedback circuits

For this reason, it was considered essential to first establish a thorough understanding of PFC operation through the study of a Single-Phase PFC converter before progressing to the more involved three-phase topology.

The single-phase PFC circuit offers a more accessible platform on which the following fundamental concepts can be examined:

- ❖ Input current shaping
- ❖ DC bus voltage regulation
- ❖ Inductor current control
- ❖ PWM switching techniques
- ❖ Closed-loop control operation

Working through the single-phase converter established a solid grounding in power circuit design as well as control implementation. The principles developed at this stage can

subsequently be extended to the three-phase PFC architecture with a well-informed perspective and greater design confidence.

Accordingly, the project work began with the simulation and analysis of a Single-Phase PFC converter, with the intention of building progressively toward more advanced three-phase topologies.

The principal PFC topologies examined and discussed in this report are as follows:

- ❖ Single-phase boost PFC topology
- ❖ Totem-Pole PFC topology
- ❖ Interleaved Totem-Pole PFC topology
- ❖ Three-Phase Diode Bridge PFC topology

4.5 Single Phase PFC Boost circuit

4.5.1 Introduction

The Single-Phase PFC converter is among the most widely deployed front-end converters in power electronic systems. Its primary purpose is to draw near-sinusoidal current from the AC supply whilst simultaneously maintaining a well-regulated DC output voltage.

Among the various single-phase PFC topologies available—including the Buck PFC, Buck-Boost PFC, and Boost PFC—the boost configuration is by far the most widely adopted, principally because:

- ❖ It provides continuous input current
- ❖ It offers better efficiency
- ❖ It enables easier current shaping
- ❖ It can generate output voltage higher than input peak voltage
- ❖ The fundamental operating principle of the converter relies on controlling the switching device in a manner that causes the input current to trace the shape of the input voltage waveform.

4.5.2 Working of Power Circuit of Single-Phase PFC

The power circuit of a typical single-phase boost PFC converter comprises an AC supply source, a full-bridge diode rectifier, a boost inductor, a power MOSFET switching device, a boost diode, and an output filter capacitor.

- ❖ Working of the Power Circuit
 - The AC supply voltage is first converted into pulsating DC by the diode bridge rectifier.
 - During the switch ON interval, energy is stored in the boost inductor.
 - When the switch turns OFF, the energy stored in the inductor is released and transferred to the output capacitor and load via the boost diode.

- Through precise PWM-based control of the switching operation, the input current waveform is shaped to closely follow the profile of the input voltage.
- The output capacitor filters the pulsating converter output, thereby maintaining a smooth and well-regulated DC bus voltage.

The inherent step-up characteristic of the boost converter topology allows the DC bus voltage to be maintained above the peak input voltage, while the current shaping action simultaneously improves the input power factor.

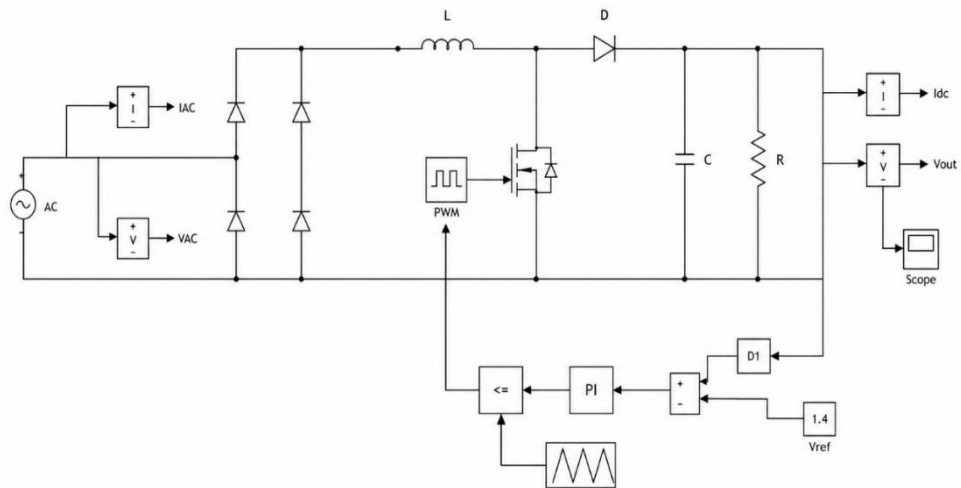


Fig. 4.1 Simulink model of Single-phase PFC circuit

4.5.3 Working of Control Circuit of Single-Phase PFC

The control circuit constitutes one of the most critical elements of the PFC converter, as it governs the switching behaviour of the power semiconductor device and, by extension, the overall performance of the converter. In broad terms, the control circuit fulfils two core functions:

- Output voltage regulation
- Input current shaping
- Voltage Control Loop

The output voltage is continuously monitored and compared against a preset reference value. Any resulting error signal is processed by a PI controller, which adjusts the converter operating point to maintain a constant DC output voltage.

- Current Control Loop

The input current reference is derived directly from the rectified input voltage waveform, ensuring the reference inherently carries the correct sinusoidal shape. The

instantaneous inductor current is then compared against this reference, and the controller adjusts the PWM duty cycle accordingly so that:

- The input current becomes sinusoidal
- The current remains in phase with the input voltage

The combined effect of these two control actions yields the following outcomes:

- Power factor improves
- Harmonics reduce significantly
- Stable converter operation is achieved

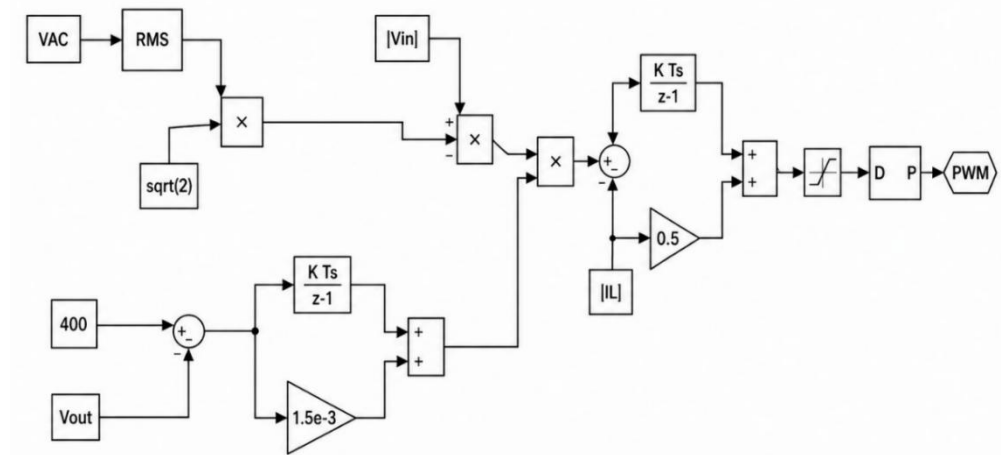


Fig. 4.2 Control Block of the Single-phase PFC simulation

4.6 Single Phase Totem-Pole PFC topology

4.6.1 Introduction

The Totem-Pole PFC topology was conceived as a direct response to the efficiency limitations imposed by the conventional diode bridge rectifier. In this configuration, the diode bridge is either partially or fully replaced by actively controlled MOSFET switches, which substantially reduces conduction losses and enhances overall conversion efficiency. As illustrated in Fig. 4.3, the Totem-Pole PFC primarily comprises a high-frequency switching leg, a low-frequency commutation leg, boost inductors, a DC bus capacitor, and the associated control circuitry.

The low-frequency leg may be realised using either diodes or MOSFETs, giving rise to two common variants: the Totem-Pole PFC with a diode low-frequency leg, and the fully active Totem-Pole PFC with a MOSFET low-frequency leg. Relative to the conventional diode bridge PFC, the Totem-Pole topology offers markedly lower conduction losses, higher efficiency, reduced thermal dissipation, greater power density,

and reduced cooling requirements—attributes that make it particularly well-suited to compact and high-efficiency wireless charging platforms.

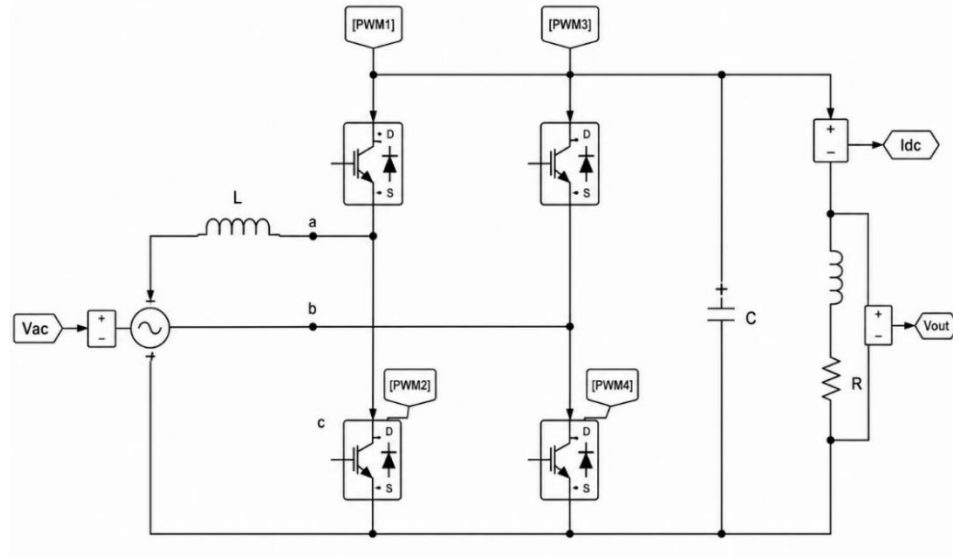


Fig. 4.3 Single-phase totem pole PFC circuit

4.6.2 Totem-Pole PFC with Diode Leg

In this variant, one bridge leg operates at high switching frequency using MOSFETs, while the complementary leg employs low-frequency diodes. The AC input current is routed through different conduction paths depending on the instantaneous polarity of the input voltage. The operating principle may be described as follows:

- ❖ During one half cycle of the AC supply, current flows through one diode conduction path in conjunction with the high-frequency MOSFET switching leg.
- ❖ During the opposite half cycle, the current is redirected through the alternate diode path.
- ❖ Throughout both half cycles, the high-frequency MOSFET switching regulates the boost inductor current and imposes the desired sinusoidal shape on the input current waveform.
- ❖ The DC bus capacitor filters the converter output and maintains a stable regulated DC voltage.

Because the diode bridge is only partially replaced, conduction losses are substantially reduced compared to the classical boost PFC. The control circuit continuously monitors the input current, output voltage, and AC supply polarity to generate the appropriate PWM drive signals for the high-frequency MOSFETs. The low-frequency diode leg conducts

passively according to the supply polarity and therefore requires no active gate drive, which simplifies the overall control implementation relative to fully active bridge configurations.

While this topology represents a meaningful improvement in efficiency over the conventional boost PFC, the diodes retained in the low-frequency conduction path continue to introduce forward voltage drops, reverse recovery losses, and associated heat generation. To eliminate these residual loss mechanisms entirely, the diode leg may be replaced with actively controlled MOSFETs, as explored in the following section.

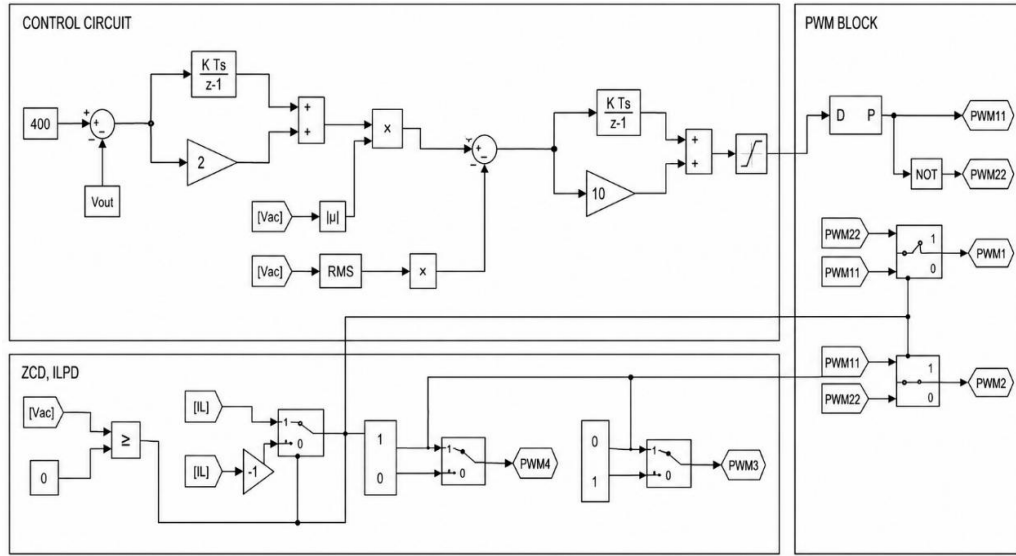


Fig. 4.4 Control circuit of Single-phase totem pole PFC circuit

4.6.3 Totem-Pole PFC with MOSFET Leg

Substituting the diode low-frequency leg with MOSFETs yields a further reduction in conduction losses, effectively eliminates reverse recovery issues, raises conversion efficiency, improves thermal performance, and enhances high-frequency switching capability—all of which make this configuration particularly well-suited to demanding high-frequency and high-power applications.

In this fully active topology, both bridge legs employ MOSFET switches. One leg operates at high switching frequency to perform current shaping, while the second leg switches at line frequency, functioning as synchronous rectifiers. The operating principle is as follows:

- ❖ During the positive half cycle of the supply voltage, one of the low-frequency MOSFETs is turned on to provide the required conduction path.
- ❖ During the negative half cycle, the complementary low-frequency MOSFET takes over the conduction.

- ❖ The high-frequency MOSFET leg handles both the current shaping function and the boost voltage conversion.
- ❖ The boost inductor alternately stores and releases energy in synchronisation with the high-frequency switching cycle.
- ❖ The output capacitor filters the pulsating inductor energy and maintains a smooth DC output voltage.

Because MOSFETs exhibit very low drain-source on-resistance compared to the forward voltage drop of rectifier diodes, conduction losses are considerably reduced. The controller must perform AC polarity detection, PWM generation, inner current loop regulation, and outer voltage loop regulation in a coordinated manner. Careful attention to gate timing is required to prevent shoot-through conditions, switching overlap, and associated current spikes. Although this topology achieves the highest efficiency of the variants studied, it does so at the cost of increased control complexity relative to the preceding configurations.

4.7 Interleaved totem pole PFC

Introduction

As power demand continues to grow, even a well-designed single-channel Totem-Pole PFC begins to exhibit elevated current stress, higher input and output ripple currents, concentrated thermal loading, and increased electromagnetic interference (EMI). The Interleaved Totem-Pole PFC topology addresses these issues by operating multiple converter phases in parallel, with their switching signals intentionally phase-shifted relative to one another.

This interleaved arrangement provides a range of performance benefits, including:

- ❖ Reduced current ripple
- ❖ Lower stress on components
- ❖ Better thermal distribution
- ❖ Reduced EMI
- ❖ Higher efficiency
- ❖ Improved dynamic response
- ❖ Smaller passive components

These advantages make it highly suitable for advanced high-power wireless charging systems.

4.7.1 Power Circuit Working

The Interleaved Totem-Pole PFC consists of multiple parallel converter channels that operate concurrently. Each phase is equipped with its own dedicated inductor and

switching devices and is driven by a phase-shifted PWM signal. The operating principle is as follows:

- ❖ The total input current is distributed across multiple converter phases, with each phase carrying a proportional share.
- ❖ The PWM switching signals for each phase are deliberately offset from one another by a fixed phase angle.
- ❖ The ripple current contributions from the individual phases partially cancel one another due to this phase offset.
- ❖ The net effect is a significant reduction in both input and output current ripple.
- ❖ Power delivery to the load becomes smoother, and stress on individual converter components is reduced accordingly.

By distributing the total current across multiple conduction paths, the topology achieves substantially higher power handling capability without proportionally increasing losses, thereby maintaining or improving overall conversion efficiency.

4.7.2 Control Circuit Working

The control circuit of the interleaved topology must manage multi-phase current balancing, PWM synchronisation, output voltage regulation, and ripple minimisation simultaneously. The controller is responsible for enforcing equal current sharing among all active phases, maintaining a stable DC output voltage, and ensuring accurate phase displacement between the individual switching signals. Precise synchronisation is critical in this topology, as any error in phase timing can result in uneven current distribution among the phases, elevated conduction losses, and degraded overall efficiency. Despite the additional control complexity that this entails, the interleaved topology delivers outstanding performance characteristics that make it highly appropriate for modern high-power wireless charging systems.

4.8 Study of Three phase PFC circuits

Introduction

As the power requirements of the wireless charging system scale upward, the inherent limitations of single-phase power conversion become increasingly apparent. High-power applications—including industrial charging infrastructure, electric vehicle charging stations, and advanced drone wireless charging systems—typically adopt three-phase power conversion owing to its capacity to manage higher power levels with improved efficiency and reduced per-device component stress.

A Three-Phase Power Factor Correction (PFC) converter is employed to transform the three-phase AC supply into a well-regulated DC voltage while simultaneously ensuring:

- High input power factor
- Low harmonic distortion
- Balanced phase currents
- High efficiency
- Stable DC bus voltage

When compared to their single-phase counterparts, three-phase PFC converters offer several notable advantages, including:

- Better power handling capability
- Reduced input current ripple
- Smaller filter requirements
- Improved efficiency
- More continuous power transfer

These attributes collectively make three-phase PFC circuits the preferred solution for high-power wireless charging applications in which stable, efficient, and continuous energy conversion is of paramount importance.

The improved performance of three-phase converters, however, comes with an associated increase in design and implementation complexity across several areas:

- Power converter topology
- Gate driving circuits
- Current sensing
- Control algorithms
- PWM generation techniques

4.9 Three-Phase Diode Bridge PFC

Introduction

The Three-Phase Diode Bridge PFC represents one of the most fundamental and widely deployed front-end converter structures in industrial power electronics. It provides a natural starting point for understanding three-phase AC-to-DC conversion principles. The topology comprises a three-phase AC supply, a six-diode bridge rectifier, boost inductors, active power switches, and a DC bus capacitor. The diode bridge rectifier first converts the three-phase AC input into an unregulated DC voltage, after which the boost converter stage applies power factor correction and regulates the output voltage to the desired setpoint.

4.9.1 Power Circuit Working

The three-phase AC input is first applied to the diode bridge rectifier, which produces a pulsating DC voltage at its output. The boost inductors and switching devices then work

in concert to shape the phase currents and regulate the output voltage. The fundamental operating sequence proceeds as follows:

- ❖ The six-diode bridge operates continuously to rectify the three-phase AC supply into a pulsating DC voltage.
- ❖ During the switch ON intervals, the boost inductors accumulate energy from the supply.
- ❖ When the switches turn OFF, the energy stored in the inductors is released and delivered to the DC bus capacitor and the connected load.
- ❖ The switching operation is controlled in a closed-loop manner to ensure that each phase current remains nearly sinusoidal and in phase with its corresponding supply voltage.
- ❖ The DC bus capacitor filters the rectified output and maintains a stable, low-ripple DC bus voltage for the downstream converter stages.

Because power is drawn from all three phases simultaneously, the instantaneous power transfer is considerably smoother and the current ripple is significantly lower than in equivalent single-phase systems.

4.9.2 Control Circuit Working

The control system incorporates voltage sensing, current sensing for all three phases, PWM signal generation, individual current control loops, and an overarching voltage regulation loop. The DC output voltage is continuously sampled and compared against a reference setpoint, with the controller adjusting the switching duty cycles to maintain a constant bus voltage. Each phase current is independently regulated to ensure it remains sinusoidal, balanced with respect to the other phases, and in phase with the corresponding phase voltage—conditions that together yield high input power factor and low harmonic distortion.

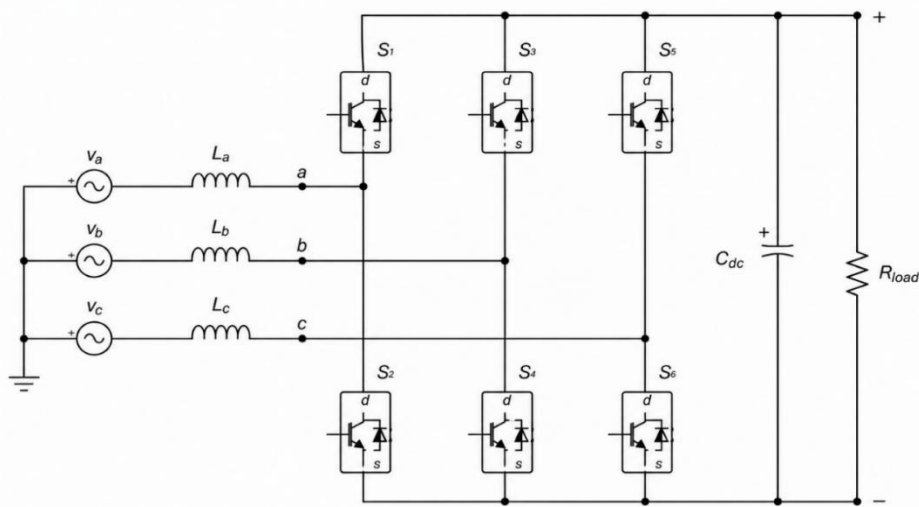


Fig. 4.5 Three-phase Diode bridge PFC circuit

4.9.3 Limitations of Three-Phase Diode Bridge PFC

Despite its simplicity and well-established reliability, the diode bridge topology is associated with several notable disadvantages:

- High conduction losses due to multiple diodes drops
- Increased heat generation
- Lower efficiency at high power levels
- Reduced power density

These drawbacks become increasingly critical in modern high-efficiency wireless charging systems, where every percentage point of power loss has a measurable impact on overall system performance and thermal management. It is precisely these limitations that have motivated the development of more advanced topologies, such as the Totem-Pole PFC described in earlier sections.

Note: Although the three-phase diode bridge PFC satisfies the basic requirement of three-phase power conversion, its high conduction losses and lower efficiency make it less suitable for high-efficiency wireless drone charging systems. A practical alternative is to realise the three-phase PFC using three independent single-phase totem pole boost PFC circuits, one for each phase. Each phase operates with its own voltage regulation and current control loop, allowing the well-understood single-phase control strategy to be directly extended to three-phase operation without requiring complex three-phase PLLs or rotating reference frame transformations. In this way, each phase performs autonomous power factor correction, and together the three converters provide a balanced and stable DC bus for the wireless charging inverter stage.

4.10 Simulation of Single-Phase Boost PFC Converter

4.10.1 Objective

The objective of this simulation is to examine the operating behavior and performance characteristics of a conventional single-phase Boost PFC converter. The simulation is intended to confirm that the input current drawn from the AC supply is successfully reshaped into a sinusoidal waveform in phase with the supply voltage, thereby achieving a near-unity power factor. In addition to verifying current waveform quality, the simulation evaluates the output voltage regulation capability of the converter

4.10.2 Output Waveforms

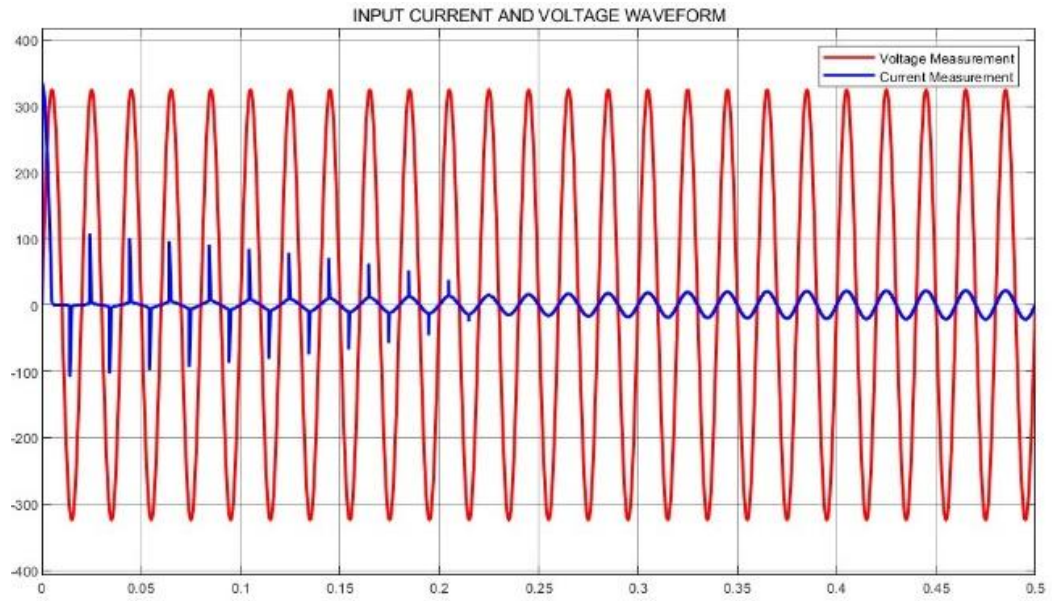


Fig. 4.6 Input Current VS Voltage waveform of Single-Phase PFC

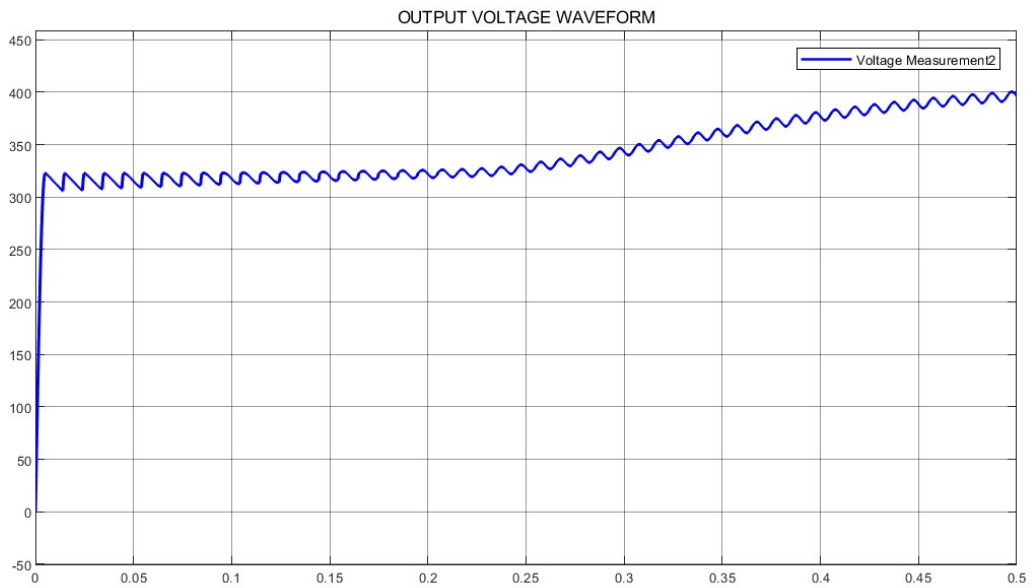


Fig. 4.7 Output Voltage Waveform of Single-Phase PFC

4.10.3 Waveform Analysis

Fig. 4.6 presents the input current and voltage waveforms of the single-phase boost PFC converter. The input current waveform tracks the sinusoidal profile of the supply voltage with good fidelity, confirming that the average current-mode controller effectively constrains the converter to draw near-sinusoidal current from the grid. The current and

voltage waveforms are substantially in phase throughout the cycle, which is consistent with a high input power factor. A degree of high-frequency switching ripple is superimposed on the current waveform as a consequence of the finite boost inductance; however, this ripple remains within acceptable bounds for grid-connected applications. The close alignment between the current and voltage envelopes provides clear validation of the correct operation of the PFC control loop.

Fig. 4.7 shows the output voltage waveform, which settles at the regulated DC bus voltage of 400 V after a brief start-up transient. A small double-line-frequency ripple component—occurring at 120 Hz for a 60 Hz supply system—is present on the output voltage waveform, which is a well-known characteristic of single-phase AC-DC conversion arising from the pulsating nature of the instantaneous AC input power. The amplitude of this ripple remains within permissible limits, confirming that the output capacitor has been adequately sized for the application. Notwithstanding these satisfactory results, the topology exhibits a fundamental limitation: the full-bridge diode rectifier places four diode voltage drops in the main conduction path, generating considerable conduction losses that detract from the overall conversion efficiency. These losses grow more significant as the current level increases, and they serve as the primary technical motivation for investigating topologies in which the rectifier diodes are partially or fully replaced by active switching devices, as undertaken in the simulations that follow.

4.11 Simulation of Single-Phase Totem-Pole PFC with low-frequency diode

4.11.1 Objective

The purpose of this simulation is to evaluate the totem-pole PFC topology in which the conventional diode bridge rectifier is replaced by a totem-pole bridge structure that retains slow-recovery diodes in the low-frequency leg while employing active MOSFETs in the high-frequency switching leg. The simulation is designed to verify enhanced power factor correction performance, quantify the reduction in conduction losses relative to the conventional boost PFC, and assess the quality of both the input current waveform and the regulated output voltage under identical operating conditions.

4.11.2 Output Waveform

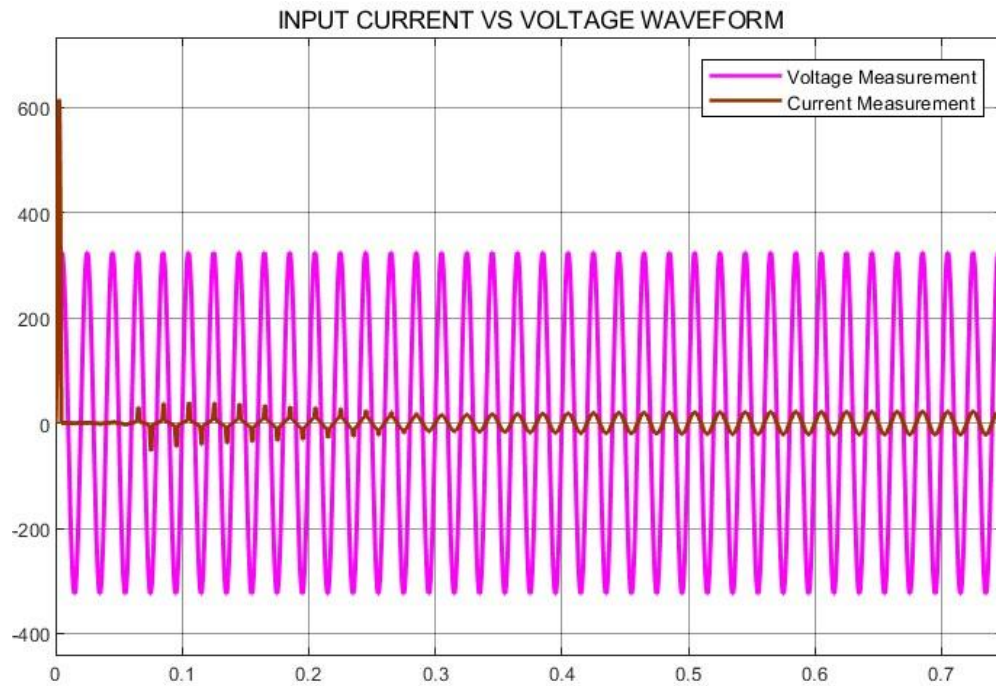


Fig. 4.8 Input Current VS Voltage waveform of Totem-Pole PFC with diodes

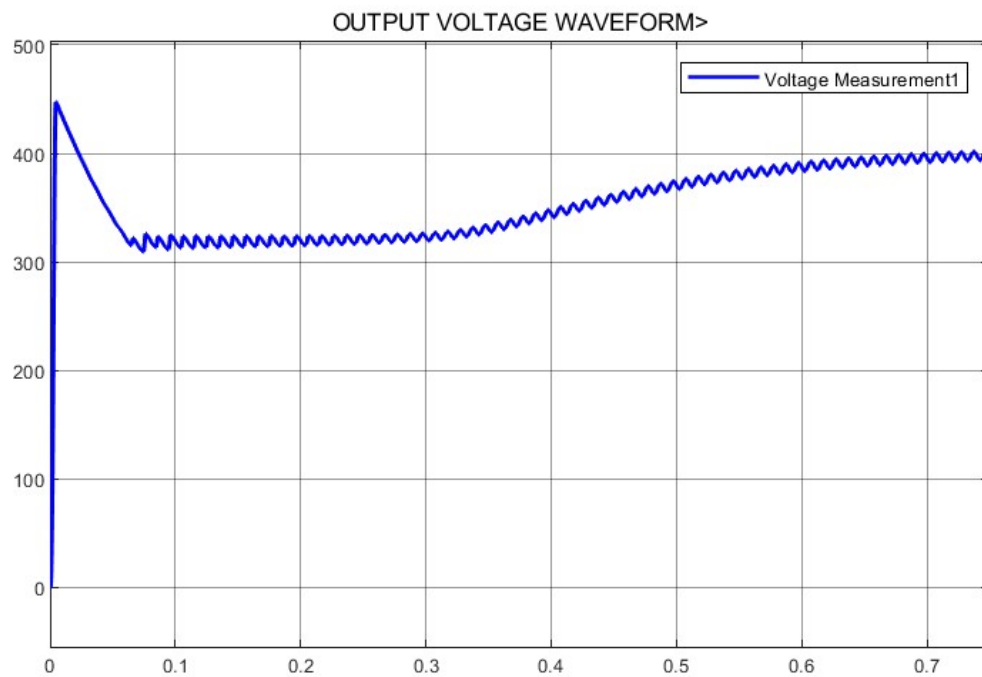


Fig. 4.9 Output Voltage Waveform of Totem-Pole PFC with diodes

4.11.3 Waveform Analysis

Fig. 4.8 presents the input current and voltage waveforms for the totem-pole PFC converter incorporating a diode low-frequency leg. The input current retains the sinusoidal profile required by the PFC controller, demonstrating good tracking of the supply voltage and confirming a high input power factor. In comparison with the conventional boost PFC result shown in Fig. 4.6, the current waveform exhibits marginally lower distortion, an improvement that can be attributed to the enhanced conduction path achieved by removing one pair of series diodes from the circuit during each half cycle. The high-frequency ripple content is broadly unchanged in magnitude, as the switching frequency and boost inductance remain at the same design values.

Fig. 4.9 shows the output voltage waveform, which stabilizes at the 400 V regulated setpoint with a start-up transient broadly comparable to that observed in the conventional boost PFC. The double-line-frequency ripple component remains present, as it is an intrinsic consequence of single-phase AC-DC power conversion and cannot be eliminated by changes to the bridge topology alone. The steady-state output voltage does, however, exhibit marginally less deviation from the reference value, which suggests a slight improvement in closed-loop response attributable to the reduced overall conduction voltage drop in the main power path. Despite the efficiency gains achieved by replacing two of the four rectifier diodes with MOSFETs in the high-frequency leg, the diodes that remain in the low-frequency conduction path continue to introduce reverse-recovery losses and finite forward voltage drops during commutation intervals. These residual losses, while smaller than in the conventional topology, ultimately constrain the peak attainable efficiency. This remaining limitation provides the rationale for the next topology, in which the low-frequency diode leg is also replaced by active MOSFETs to realise truly synchronous rectification and eliminate all significant conduction-related loss mechanisms.

4.12 Simulation of Single-Phase Totem-Pole PFC with MOSFET Low-Frequency Leg

4.12.1 Objective

This simulation is made to characterize the fully active totem-pole PFC converter, in which both the high-frequency switching leg and the low-frequency commutation leg employ MOSFET devices in place of diodes. The simulation is designed to demonstrate a further improvement in both conversion efficiency and input current quality, achieve a power factor approaching unity, and confirm stable output voltage regulation under steady-state loading. It is anticipated that the results will validate the fully active MOSFET totem-pole topology as the most power-dense and highest-efficiency architecture among the three PFC converters investigated, making it especially well-suited to the compact form factor and demanding performance requirements of wireless charging applications.

4.12.2 Output Waveforms

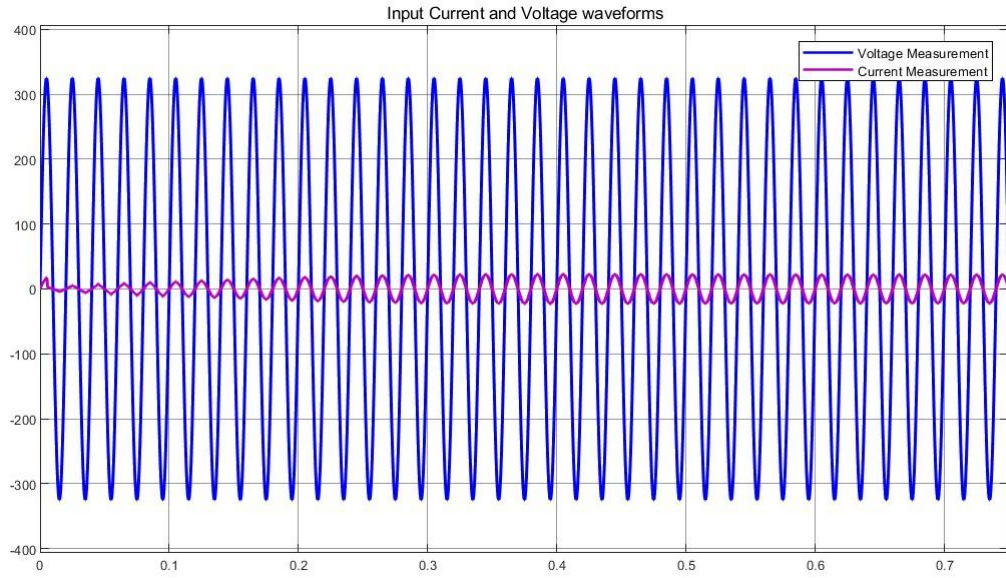


Fig. 4.10 Input Current VS Voltage waveform of Totem-Pole PFC with Mosfets

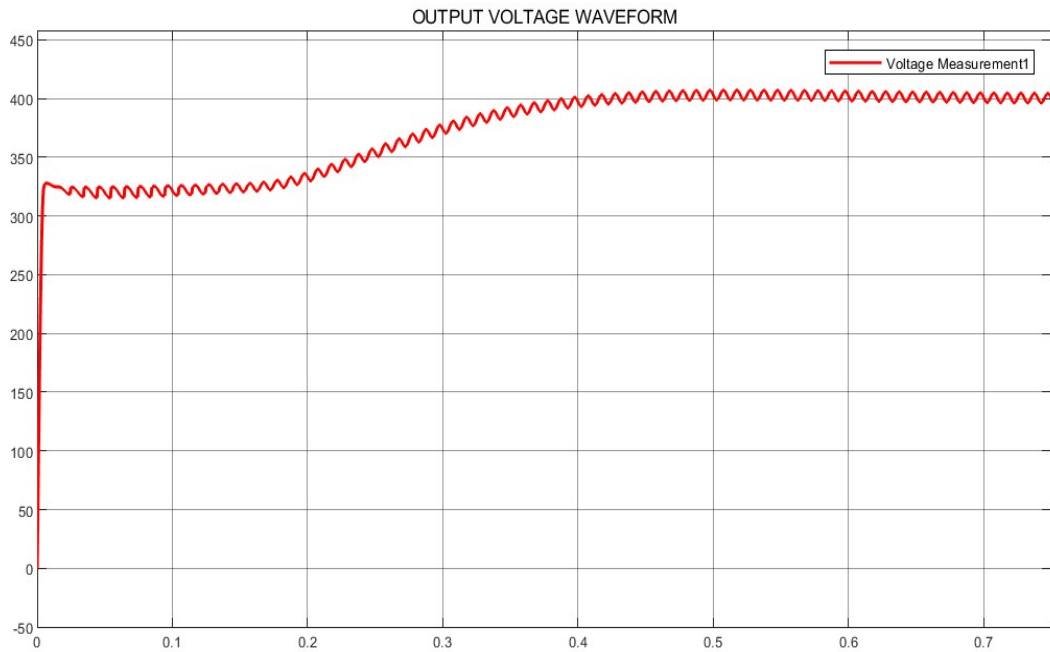


Fig. 4.11 Output Voltage Waveform of Totem-Pole PFC with Mosfets

4.12.3 Waveform Analysis

Fig. 4.10 presents the input current and voltage waveforms for the fully active totem-pole PFC converter, in which MOSFETs are employed in both the high-frequency and low-frequency bridge legs. Among all three topologies examined in this study, the

input current waveform produced by this configuration most closely approximates a pure sinusoid, exhibiting minimal harmonic distortion and negligible dead-zone distortion in the vicinity of the supply voltage zero-crossing. The current remains tightly in phase with the voltage throughout the complete cycle, yielding a power factor that approaches unity. The elimination of reverse-recovery transients from the low-frequency conduction path—a source of current spikes that affected the previous topology at input voltage polarity transitions—is clearly visible in the improved zero-crossing behaviour of the current waveform. This enhancement is a direct result of the active MOSFET commutation strategy employed in the low-frequency leg.

Fig. 4.11 shows the output voltage waveform for this topology, which converges to the 400 V setpoint more rapidly than in the earlier topologies, with a noticeably shorter settling time. The amplitude of the double-line-frequency ripple is marginally lower than in the preceding cases, which is consistent with the higher conversion efficiency and reduced internal power dissipation of this configuration. The steady-state output voltage exhibits excellent regulation, confirming the effectiveness of the voltage control loop under resistive loading conditions. In summary, the fully active totem-pole topology achieves the highest conversion efficiency, the lowest total harmonic distortion (THD) in the input current, and the best output voltage regulation of the three PFC converters examined in this study. Although the control complexity and gate-drive circuit requirements are greater than those of the diode-leg variant, the performance advantages clearly justify this trade-off, particularly in the context of high-power wireless charging systems where compact packaging and minimal thermal management overhead are important design constraints.

4.13 Analysis with RL-loads

When the PFC converter is operated into an RL load rather than a purely resistive termination, the converter behaviour is modified in several noteworthy respects. The inductive component of the load introduces a phase lag between the load terminal voltage and the load current, which alters the instantaneous power demand presented to the output capacitor and consequently modifies the dynamic behaviour of the voltage control loop. Under RL loading conditions, the output voltage ripple amplitude may increase marginally, since the reactive energy cycling between the output stage and the load inductance generates additional perturbations that the output capacitor must absorb, and the lagging load current means that the capacitor voltage is more sensitive to transient changes in the demanded power.

Under RL loading, the input current waveform remains nearly sinusoidal, showing that the inner current control loop of the PFC converter responds quickly enough to handle slower load dynamics without affecting current quality. The power factor also remains high, with only slight degradation at lighter RL loads due to increased current ripple. Output voltage regulation maintains good steady-state accuracy, although transient overshoot and settling time are slightly higher compared to purely resistive loads because

of the energy stored in the load inductance. Overall, these results demonstrate the robustness and practical suitability of the totem-pole PFC topology for inductive applications such as wireless charging systems and motor drive front ends.

4.14 Comparative Analysis of Simulated PFC Topologies

A structured performance comparison of the three PFC topologies was carried out, with evaluation criteria encompassing conversion efficiency, conduction losses, switching performance, achievable power factor, and suitability for wireless charging applications.

Table 4.1: Performance Comparison of Different PFC Topologies

Parameter	Single-Phase Boost PFC	Totem-Pole with Diode Leg	Totem-Pole with MOSFET Leg
Conduction Losses	High	Moderate	Very Low
Efficiency	Moderate	High	Very High
Power Factor	High	Higher	Closest to Unity
Switching Frequency Capability	Moderate	High	Very High
Thermal Performance	Moderate	Better	Excellent
Ripple Reduction	Moderate	Moderate	Good
Control Complexity	Simple	Moderate	High
Suitability for Wireless Charging	Moderate	Good	Excellent

CHAPTER – 5

INTRODUCTION TO HIGH FREQUENCY INVERTER

5.1 High Frequency Inverter in Wireless Power Transfer

A High Frequency Inverter (HFI) is a crucial power electronic component in Wireless Power Transfer (WPT) systems, responsible for converting a regulated DC input into a high-frequency AC output. This high-frequency AC is necessary to generate the alternating magnetic fields required for efficient wireless energy transfer. In a typical WPT system, the inverter is placed between the DC source and the resonant network, supplying high-frequency excitation to the transmitting coil.

WPT technology integrates multiple domains such as power electronics, electromagnetics, and control systems to enable contactless energy transfer. Among these, the inverter plays a central role because the efficiency and performance of the entire system heavily depend on the quality and frequency of the AC power supplied to the coupling coils. Modern WPT systems are increasingly moving toward higher operating frequencies to achieve system miniaturization, higher power density, and improved efficiency.

High-frequency operation allows the use of smaller inductors and capacitors in the resonant network, reducing the overall size and weight of the system. However, it also introduces significant challenges such as increased switching losses, effects of parasitic elements, electromagnetic interference (EMI), and higher driving complexity of switching devices. These factors make the design of high-frequency inverters more critical and complex compared to conventional low-frequency inverters.

5.2 Selection of 85 kHz operating frequency

The operating frequency is a critical design parameter in Wireless Power Transfer (WPT) systems, as it directly influences system efficiency, size of components, power density, and overall performance. In high-frequency WPT systems, increasing the operating frequency allows for reduced size of inductive and capacitive components, enabling compact and lightweight designs. This trend toward higher frequency operation is driven by the need for miniaturization and improved power density in modern applications.

However, the selection of frequency involves important trade-offs. At lower frequencies, larger coils and bulky compensation components are required, resulting in increased system size and reduced power density. On the other hand, very high frequencies (in the MHz range) introduce significant switching losses in power electronic devices, increased electromagnetic interference (EMI), and challenges related to parasitic elements and device limitations. These factors restrict the practical operating range of WPT systems.

In this project, an operating frequency of 85 kHz is selected as it provides an optimal balance between efficiency, component size, and system reliability. This frequency lies within the widely adopted range for inductive wireless power transfer systems, particularly in applications such as electric vehicle charging and industrial systems. Operating at 85 kHz allows the use of reasonably sized coils while maintaining high coupling efficiency and manageable switching losses.

Additionally, at 85 kHz, modern power electronic devices such as MOSFETs can operate efficiently with acceptable switching losses and thermal performance. It also simplifies the design of resonant compensation networks and reduces the impact of parasitic capacitances and inductances compared to higher frequency systems. Furthermore, this frequency range ensures compliance with electromagnetic compatibility (EMC) and safety standards.

5.3 Classification of inverters in WPT

5.3.1 Inverter Topologies in WPT system

In Wireless Power Transfer (WPT) systems, inverters are classified based on their operating principles and circuit configurations. As shown in Fig. 5.1, inverters used in WPT systems can be broadly divided into three major categories: Oscillator-based inverters, Power Amplifier-based inverters, and H-Bridge inverter topologies with their derived forms.

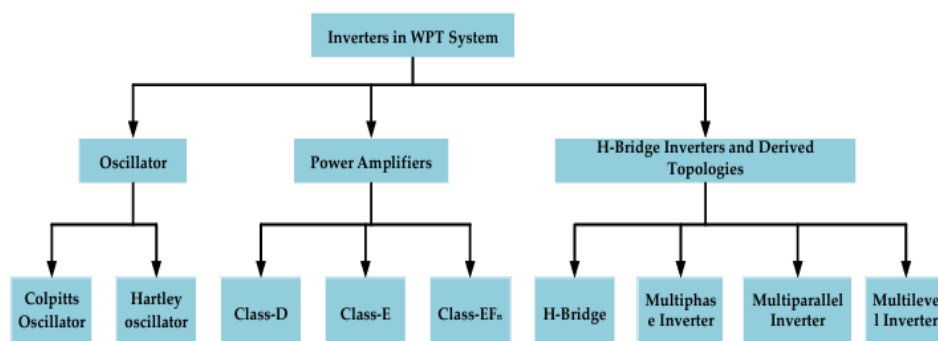


Fig 5.1 Classification of inverters in the WPT system

1. Oscillator-Based Inverters

Oscillator-based inverters are the simplest type of high-frequency inverter, where oscillations are generated using passive LC components without the need for complex switching control. These circuits rely on the natural energy exchange between inductors and capacitors to produce high-frequency AC signals.

The most common oscillator circuits used in WPT systems are:

- Colpitts Oscillator: Uses a capacitive voltage divider for feedback and provides relatively stable high-frequency oscillations. However, its frequency control is limited and it is sensitive to component variations.
- Hartley Oscillator: Utilizes an inductive feedback network and is simpler in structure, but it produces higher harmonic distortion and less stable output compared to Colpitts oscillators.

Although oscillator-based inverters offer simple design and high-frequency operation, they are rarely used in modern WPT systems due to poor controllability, low efficiency, and high harmonic content.

2. Power Amplifier-Based Inverters

Power amplifier-based inverters are widely used in high-frequency WPT systems, especially for low- to medium-power applications. These inverters operate in switching mode to improve efficiency by minimizing overlap between voltage and current in switching devices.

The main types include:

- Class-D Amplifier: Uses two switching devices operating alternately to produce a square-wave output. It offers simple structure and high efficiency but may introduce DC components and requires proper compensation.
- Class-E Amplifier: A single-switch topology known for achieving soft-switching conditions such as Zero Voltage Switching (ZVS), resulting in very high efficiency. However, it experiences high voltage stress across the switching device and is sensitive to load variations.
- Class-E_{F_n} (or Class-Φ_n) Amplifier: An improved version of Class-E that suppresses harmonics using resonant networks, leading to better efficiency and reduced electromagnetic interference (EMI).

These amplifiers are particularly suitable for high-frequency (MHz range) applications but are generally limited in power handling capability and require precise tuning of resonant circuits.

3. H-Bridge Inverters and Derived Topologies

H-Bridge inverters are the most commonly used inverter topology in medium- and high-power WPT systems due to their robustness, controllability, and scalability. The basic H-bridge consists of four switching devices arranged in two bridge arms, allowing conversion of DC into high-frequency AC.

Based on system requirements, several derived topologies are used:

- H-Bridge Inverter: The standard full-bridge configuration that generates a square-wave AC output. It offers high efficiency, simple control, and is widely used in practical WPT systems.
- Multiphase Inverter: Uses multiple inverter phases operating with phase shifts to reduce current ripple, improve power sharing, and enhance system efficiency. It is suitable for high-power applications.
- Multi-parallel Inverter: Consists of multiple inverter units connected in parallel to distribute current stress across devices. This improves thermal performance and increases power handling capability.
- Multilevel Inverter: Generates stepped AC waveforms with multiple voltage levels, reducing harmonic distortion and improving output waveform quality. It is beneficial in high-power and high-voltage applications.

5.3.2 Voltage Source and Current Source Inverters (H-Bridge Based)

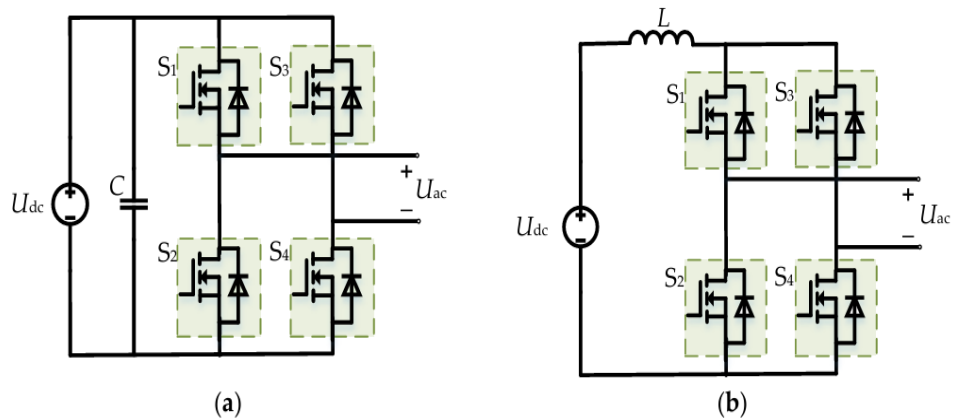


Fig 5.2 a) Voltage Source Inverter b) Current Source Inverter

A Voltage Source Inverter (VSI) is an H-bridge inverter in which the input is a constant DC voltage, typically maintained using a DC-link capacitor. The inverter converts this DC

voltage into a high-frequency AC voltage by alternately switching the diagonal pairs of switches in the H-bridge configuration. The output is generally a square-wave voltage, which is then applied to the resonant network of the WPT system. VSI-based inverters are widely used due to their simple control strategy, high efficiency, and ease of implementing soft-switching techniques. They also offer good load adaptability and are suitable for medium- and high-power WPT applications.

A Current Source Inverter (CSI) is an H-bridge inverter where the input is a constant DC current, typically maintained using a large series inductor. The inverter generates a high-frequency AC current output by controlling the switching sequence of the H-bridge devices. Unlike VSI, the output is a square-wave current, which drives the resonant network. CSI-based inverters offer advantages such as inherent short-circuit protection and robust current control. However, they require bulky inductors, have more complex implementation, and do not tolerate open-circuit conditions, which limits their practical use in many WPT systems.

5.4 Voltage Source H – Bridge Inverter

Having established a stable, regulated DC bus at the output of the PFC rectifier stage, the next functional block in the WPT system is the high-frequency inverter. The regulated 400 V DC output from the PFC stage serves as the input to the H-bridge inverter, which is responsible for converting this constant DC voltage into a high-frequency AC signal suitable for exciting the transmitter coil of the wireless power transfer system.

5.4.1 Architecture of Voltage Source H–Bridge Inverter

The inverter is built around a full-bridge (H-bridge) topology, which was chosen for this design owing to its well-established performance at medium-to-high power levels and its ability to produce a full bipolar square-wave output without the need for a transformer. The circuit, as implemented in the MATLAB/Simulink environment and shown in Figure 5.3, consists of four MOSFET switches — designated H1, H2, H3, and H4 — arranged in two bridge legs. Each switching device is paired with an anti-parallel freewheeling diode, which provides a current path during the intervals when the complementary switches are off and ensures safe commutation during high-frequency transitions. Gate drive signals are generated externally and supplied to the switches in two complementary pairs, labelled [A] and [B] in the Simulink model.

5.4.2 Working Principle of VSI

The H-bridge operates by alternately enabling two diagonal switch pairs. In the first half-cycle, switches H1 and H4 (controlled by signal [A]) are turned ON simultaneously, connecting the load between the positive DC rail and ground and

producing a positive output voltage of approximately +400 V across the resonant network. In the second half-cycle, switches H2 and H3 (controlled by signal [B]) are activated, reversing the polarity across the load and producing an output of -400 V. By repeating this switching sequence at 85 kHz, the inverter generates a symmetrical square-wave AC output at the desired operating frequency.

The rated output of the inverter stage under these conditions is 400 V AC at a coil current of 5 A, yielding a delivered power of approximately 2 kW to the transmitter side.

5.5 Simulation and Results

The proposed Voltage Source H-Bridge Inverter was simulated to evaluate its performance in the Wireless Power Transfer (WPT) system. The inverter circuit consists of four switches arranged in an H-bridge configuration and is driven by complementary gate signals to generate a high-frequency AC output from a DC input source.

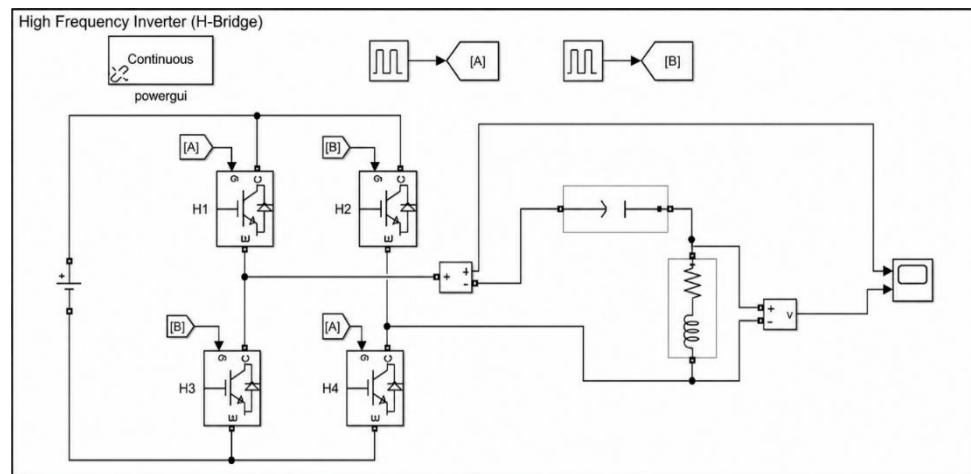


Fig 5.3 High Frequency H-Bridge Inverter Simulation Model

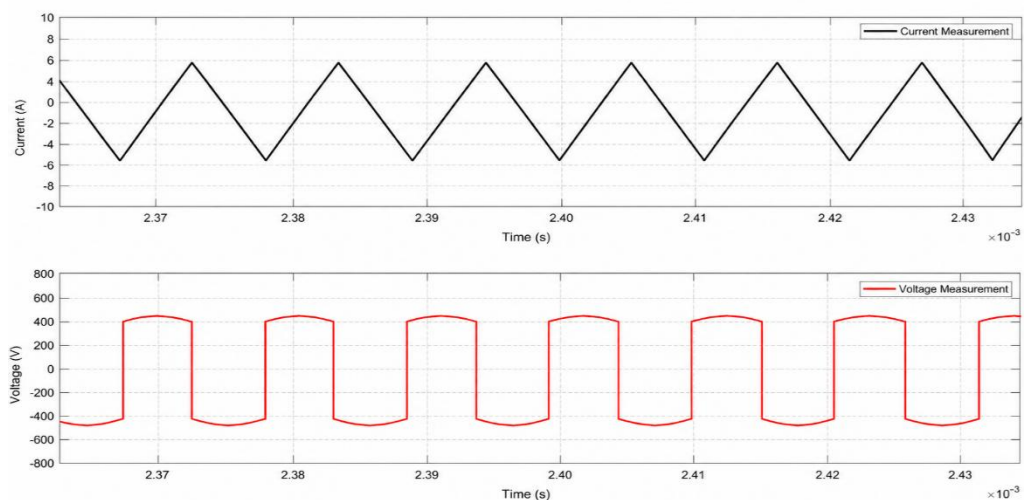


Fig 5.4 Voltage and Current Waveforms of H-Bridge Inverter

Circuit Parameters

Table 5.1 Values of circuit parameters

Parameter	Symbol	Value
Output Voltage	V_{out}	400 V
Output current	I_{out}	5 A
Resistance	R	$10\ \Omega$
Inductance	L	$120\ \mu H$
Capacitance	C	100 nF

The simulation results confirm that the H-bridge inverter operates in close agreement with the theoretical expectations outlined in the preceding sections. As observed from the output voltage waveform, the inverter produces a well-defined square wave, which is the expected behaviour when diagonal MOSFET pairs switch alternately at the designed frequency of 85 kHz. The transitions between the positive and negative voltage levels are sharp and consistent, indicating stable gate switching and the absence of any significant timing asymmetry between the two bridge legs.

CHAPTER 6

COIL DESIGN AND CONFIGURATION

6.1 Introduction

In any wireless power transfer (WPT) system, the coils form the heart of the energy exchange process. The transmitting coil, energised by high-frequency alternating current from the H-Bridge inverter, generates an oscillating magnetic field. The receiving coil intercepts this field and converts it back into usable electrical energy through electromagnetic induction. The efficiency, reliability, and practical performance of the entire wireless charging system therefore depend heavily on the geometry, material, and arrangement of these coils. This chapter presents the rationale for choosing circular coil geometry, examines the available coil configurations, and concludes with a detailed comparative analysis of the 1:1 and 1:4 configurations studied in this project.

6.2 Why Circular Coils?

The choice of coil geometry is not arbitrary. Several coil shapes circular spiral, square or rectangular spiral, and Double-D (DD/DDQ) are used across WPT applications, each with distinct electromagnetic characteristics. For this agricultural drone charging system, concentric circular coils were selected as the preferred geometry for the following reasons.

Axial Symmetry and Predictable Coupling: Circular coils possess complete rotational symmetry about their central axis. For a coaxial transmitter-receiver arrangement, this symmetry maximises mutual inductance for a given coil radius and air gap when compared to other shapes. It also simplifies analytical design since closed-form and semi-analytical expressions exist for calculating self-inductance and mutual inductance of concentric circular turns, making it straightforward to predict performance and tune resonance before physical prototyping.

High Quality Factor: The quality factor (Q) of a coil directly governs WPT efficiency. Optimisation studies on spiral coils modelled as sets of concentric circular turns consistently show that appropriate selection of inner radius, outer radius, and turn spacing maximises Q . A higher Q coil stores more reactive energy per unit resistive loss, which translates into better power transfer efficiency at resonance.

Compatibility with Drone Geometry: Agricultural drones and UAVs typically have a circular or radially symmetric undercarriage. Circular receiving coils integrate naturally

with this geometry, taking advantage of radial symmetry during landing. This also simplifies the mechanical mounting of the receiver coils on the drone's body.

Scalability to Multi-Coil Structures: Circular coils can be extended into decoupled multi-coil WPT systems with controlled coupling between inner and outer coils. This property is particularly useful when scaling from a simple 1:1 configuration to the 1:4 arrangement used in this project, as the individual receiver coils can be uniformly distributed around the drone's frame while preserving symmetry.

In comparison, square coils better utilise rectangular PCB areas but offer uneven field distribution, while DD and DDQ coils though highly tolerant to lateral misalignment introduce structural complexity and are primarily suited to high-power stationary EV charging platforms rather than compact, distributed drone applications.

6.3 Coil Configurations in WPT Systems

The relative number and arrangement of transmitting (Tx) and receiving (Rx) coils defines the coil configuration, and this choice significantly impacts power capture, alignment tolerance, voltage levels, and system complexity.

6.3.1 1:1 Configuration - Single Transmitter, Single Receiver

This is the simplest and most widely studied arrangement. One transmitting coil on the charging pad couples magnetically with one receiving coil mounted on the drone. It offers ease of design and straightforward resonance tuning. However, the 1:1 configuration is highly sensitive to alignment: any lateral or angular misalignment between the two coils rapidly degrades the coupling coefficient and reduces power transfer efficiency. It is best suited for controlled environments where precise landing can be guaranteed.

6.3.2 1:N Configuration - Single Transmitter, Four Receivers

A single large transmitting coil on the charging pad couples with N smaller receiving coils distributed across the drone's undercarriage. The N receiver coils are connected in series, summing their individually induced voltages to meet the drone's required input voltage. This configuration significantly improves alignment tolerance, since even when the drone lands with some offset, at least a subset of the receiver coils remains within effective coupling range. It also distributes induced current across four coils, reducing resistive heating in each coil individually.

6.3.3 N:1 Configuration - Multiple Transmitters, Single Receiver

Here, multiple transmitting coils on the ground pad energise a single receiver coil on the drone. The distributed transmitter arrangement creates a larger effective charging zone on the pad surface, making this configuration tolerant to variations in where the drone lands. It is useful in scenarios where the drone's landing position cannot be precisely controlled, though it requires more complex transmitter-side power electronics and synchronisation.

6.3.4 N:N Configuration - Multiple Transmitters, Multiple Receivers

The most flexible arrangement, where several transmitter coils pair with several receiver coils simultaneously. This supports multi-device charging and entirely free-position placement, meaning the user or in this case, the drone need not align with any specific region of the charging pad. While this offers maximum versatility, it comes at the cost of significantly higher system complexity, increased cross-coupling interference between adjacent coils, and more elaborate power management circuitry.

6.4 Selected Configurations

Based on the system requirements: a 2-kW power rating, 85 kHz operating frequency, 48 V / 40 A charging output, and operation in unstructured agricultural environments three configurations were studied and simulated in this project: the 1:1 configuration as a baseline, the 1:2 configuration as an intermediate step, and the 1:4 configuration as the proposed system. All three were modelled electromagnetically in Ansys Maxwell 3D and validated at the circuit level in Ansys Twin Builder.

6.4.1 Coil Parameters

The operating frequency was fixed at 85 kHz, consistent with international WPT standards for high-power applications. This frequency provides an optimal trade-off: it is high enough to allow compact coil dimensions and reduced material usage approximately 50% less copper compared to 22 kHz systems while remaining low enough to ensure efficient switching in modern SiC MOSFETs with manageable thermal dissipation. Coil turn calculations were performed analytically:

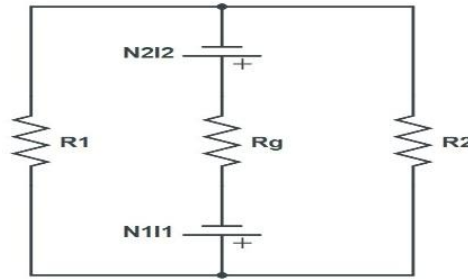


Fig 6.1 Circuitual representation of the Coil system

$$\phi = \frac{N_1 I_1 - N_2 I_2}{R_g} \quad \text{-----6.4.1.1}$$

$$E_{1,rms} = N_1 \phi_{rms} \omega \quad \text{-----6.4.1.2}$$

$$E_{2,rms} = N_2 \phi_{rms} \omega \quad \text{-----6.4.1.3}$$

With the operating parameters inter-coil distance $d = 3.5$ cm, mean coil separation $D = 10$ cm, induced receiver voltage $E_2 = 400$ V, and $\omega = 2\pi f$ the number of turns for both transmitting and receiving coils was determined to be 23 turns. This theoretical value served as the starting point for electromagnetic simulation, where it was further refined based on observed flux linkage and induced voltage results.

6.5 Electromagnetic Simulation

Both configurations were simulated using the **Transient Magnetic Solver** in Ansys Maxwell 3D. The transmitting coil was excited with a 5 Ampere AC current in all cases. Magnetic flux distribution, mutual inductances, and induced voltages were computed for each configuration. The extracted coil parameters self-inductances, series resistances, and mutual inductance matrices were subsequently imported into Ansys Twin Builder for circuit-level analysis of power transfer efficiency across the 80–90 kHz frequency range.

6.5.1 1:1 Configuration Simulation

The 1:1 model comprised a single transmitting coil and a single receiving coil placed coaxially at the specified air gap of 3.5 cm, with a mean coil separation of 10 cm. The transmitter was excited with a 5 Ampere AC current, consistent across all configurations studied in this project. The Maxwell 3D simulation confirmed effective and concentrated magnetic coupling between the coil pair, with the flux distribution remaining tightly symmetric about the shared coil axis. Since both coils share the same central axis and geometry, the flux linkage is maximised under this perfectly aligned condition, and the magnetic field vector plot shows clean, closed flux paths connecting the transmitter and receiver with minimal leakage. The extracted coil parameters self-inductance, series resistance, and mutual inductance were subsequently imported into Ansys Twin Builder, where the complete circuit model was assembled with series compensation capacitors on both transmitter and receiver sides, and a resistive load representing the rectifier input stage.

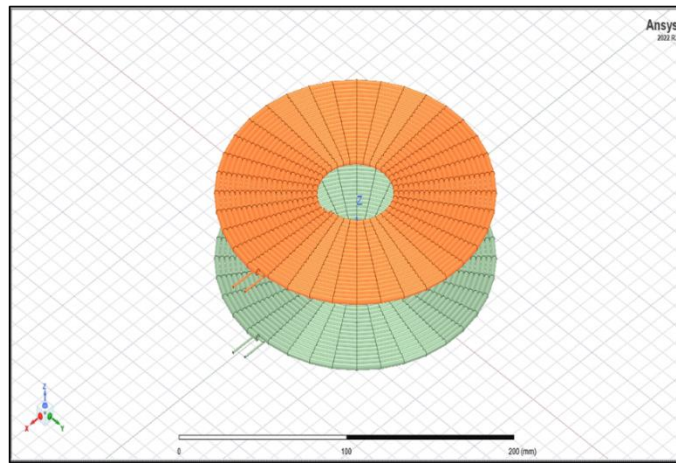


Fig. 6.2 Single Coil Simulation in Ansys Maxwell 3D

The Twin Builder frequency sweep across the 80–90 kHz band produced a sharp, narrow resonance peak in the efficiency curve, reaching a maximum efficiency of **95.57% at exactly 85.0 kHz**. The sharpness of this peak is characteristic of a tightly coupled, lightly loaded single-receiver system with a high quality factor — the system stores and transfers reactive energy very efficiently at resonance, but efficiency degrades steeply on either side of the resonant point. This narrow bandwidth, while delivering the highest peak efficiency among all three configurations studied, also makes the 1:1 system sensitive to any deviation from the nominal operating frequency caused by component tolerances or temperature-dependent inductance variation in a real hardware implementation.

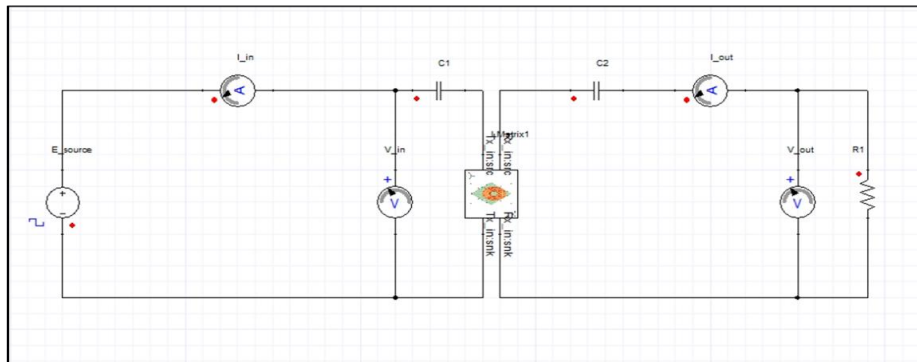


Fig 6.3 Wireless Power transfer simulation in Twin Builder

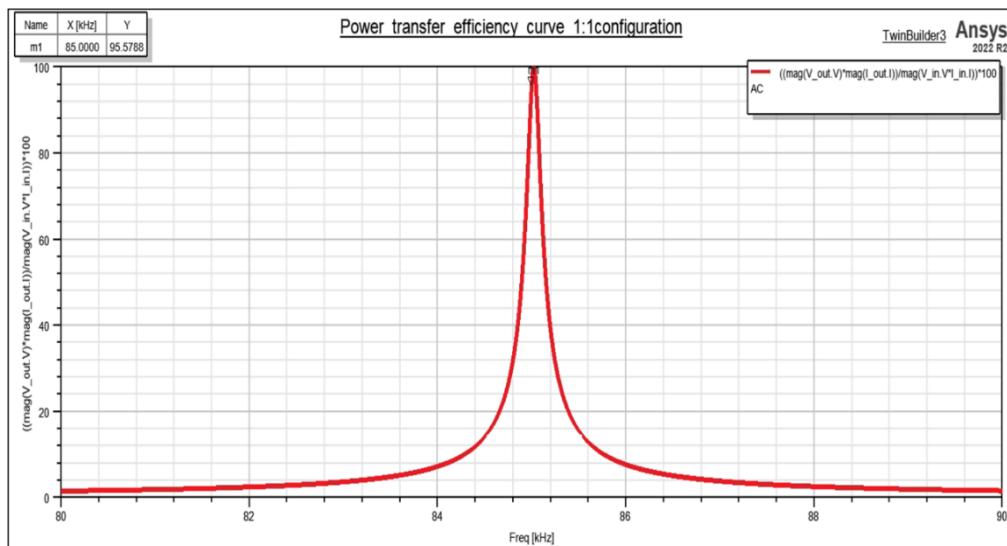


Fig 6.4 Power transfer efficiency curve for 1:1 configuration

6.5.2 1:2 Configuration Simulation

The 1:2 model placed one transmitting coil and two receiving coils in a coplanar arrangement, with the two receivers positioned symmetrically on either side of the transmitter axis. The transmitter was again excited at 5 A AC, consistent with the other configurations. The Maxwell 3D simulation confirmed successful magnetic coupling between the single transmitter and both receiver coils, with the flux distribution showing effective linkage across both receivers while exhibiting the expected attenuation with lateral distance from the transmitter centre.

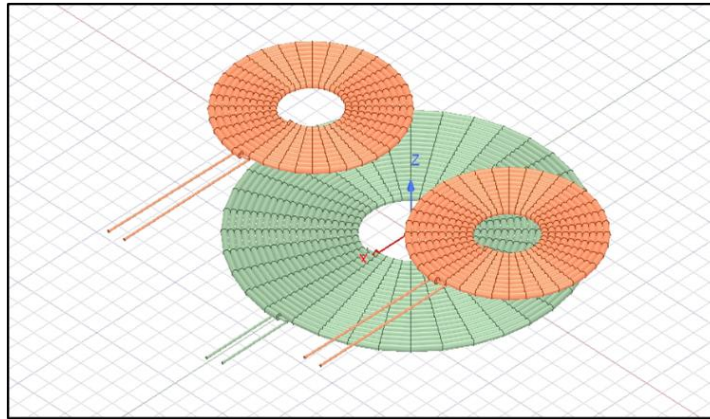


Fig 6.5 Coil Simulation in Ansys Maxwell 3D

The Twin Builder frequency sweep across the 80–90 kHz band produced an efficiency curve that peaked at **94.79% at 84.97 kHz**, showing a broader resonance profile compared to the sharp, narrow peak observed in the 1:1 configuration. This broadening is a direct consequence of the increased loading on the transmitter side due to the additional receiver coil, which slightly shifts the effective resonant point and distributes the reactive energy across a wider frequency range. The result is a system that is somewhat more tolerant to small deviations from the nominal operating frequency, which is practically beneficial in real hardware implementations where component tolerances can cause slight frequency drift.

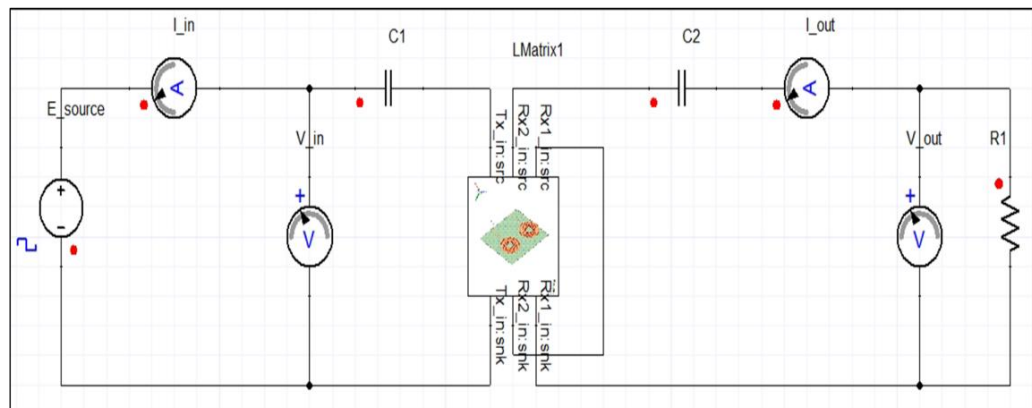


Fig 6.6 Wireless Power transfer simulation in Twin Builder

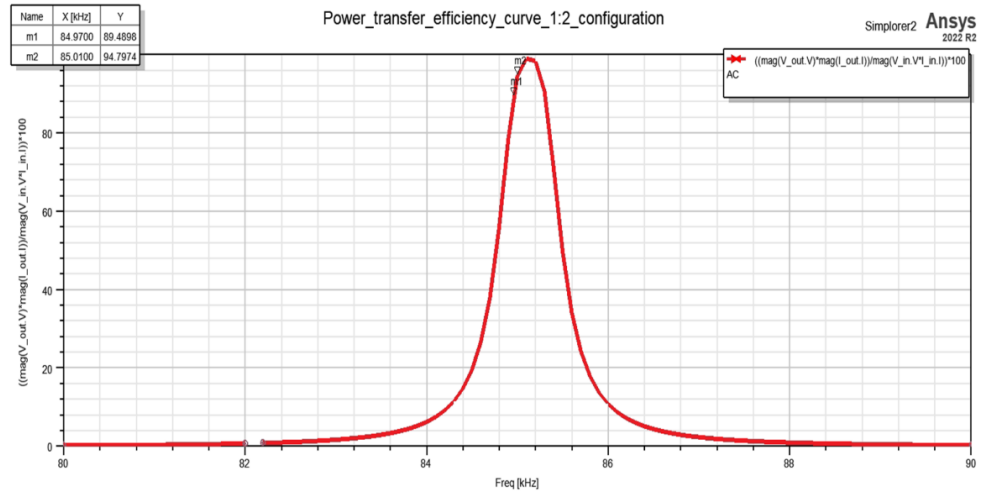


Fig 6.7 Power transfer efficiency curve for 1:1 configuration

6.5.3 1:4 Configuration Simulation

The 1:4 model placed one central transmitting coil on the ground pad and four receiving coils arranged at the corners of a $50 \text{ cm} \times 50 \text{ cm}$ square, with the centre-to-centre distance between adjacent receiver coils fixed at 50 cm to accurately reflect the physical dimensions of the target drone platform. Each receiver coil was positioned to replicate the spatial arrangement of the drone's landing legs, and all four receiver coils were connected in series in the circuit model. The transmitter was excited at 5 A AC , consistent with the other configurations. The Maxwell 3D simulation using the Transient Magnetic Solver computed the magnetic field distribution, mutual inductances between the transmitter and each of the four receivers, and the induced voltages across the receiver array. The colour-mapped flux density plot showed a maximum flux density of approximately $2574 \mu\text{T}$ in the vicinity of the transmitter, with a gradual and symmetric attenuation of field strength outward toward the four distributed receiver coils. Importantly, the magnetic field vector plot confirmed the formation of closed flux paths linking the central transmitter with all four surrounding receivers, with the coupling distribution remaining relatively symmetric across the array despite the spatial separation between coils. The extracted inductance matrix, containing the self-inductances of all five coils and the mutual inductances between every transmitter-receiver pair, was imported into Ansys Twin Builder for circuit-level co-simulation.

The Twin Builder frequency sweep across the $80\text{--}90 \text{ kHz}$ band yielded a peak efficiency of **94.79% at 85.01 kHz**, with the resonance curve exhibiting a broader profile compared to the sharp peak observed in the 1:1 configuration. This broadening arises from the increased effective loading presented by the four series-connected receiver coils to the transmitter, which distributes the reactive energy exchange across a slightly wider frequency range and makes the system less sensitive to small deviations from the nominal resonant frequency. The induced voltages across the four individual receiver coils, when summed through the series connection, successfully approached the target rectifier input

of 400 V, validating the series voltage addition principle as a practical means of meeting the drone's charging voltage requirement without demanding impractically high flux linkage from any single small receiver coil.

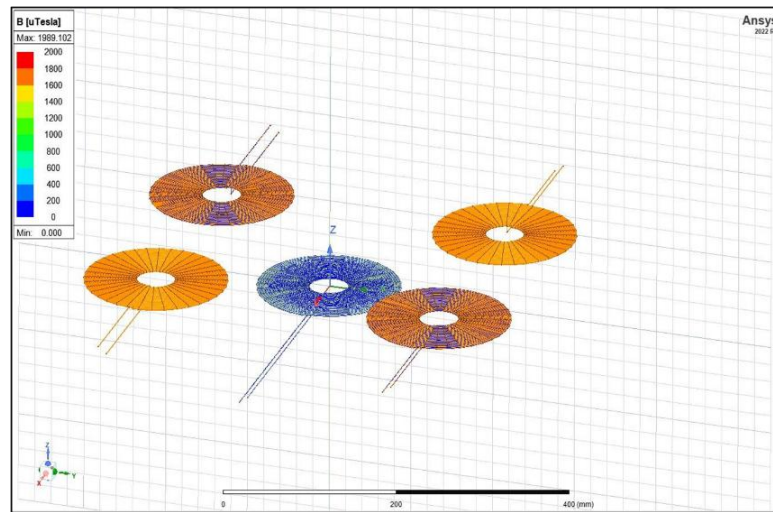


Fig 6.8 Coil Simulation in Ansys Maxwell 3D

6.6 Comparative Analysis: 1:1, 1:2, and 1:4 Configurations

The following table summarises the key parameters obtained from simulation across all three configurations.

Table 6.1 Comparative analysis of various configurations

Parameter	1:1 Configuration	1:2 Configuration	1:4 Configuration
Number of Tx Coils	1	1	1
Number of Rx Coils	1	2 (series)	4 (series)
Resonant Frequency	85.0 kHz	84.97 kHz	85.01 kHz
Peak Efficiency	95.57%	94.79%	94.79%
Effective Charging Area	Small	Moderate	Large
Misalignment Tolerance	Low	Moderate	High
Induced Voltage per Rx Coil	Full output	Shared across 2	Shared across 4
Resistive Losses per Coil	Concentrated	Divided across 2	Divided across 4
Resonance Bandwidth	Narrow	Broader	Broader
System Complexity	Low	Low–Moderate	Moderate
Suitability for Agriculture	Limited	Partial	High

DISCUSSION

The three configurations reveal a consistent and informative trend as the number of receiver coils is increased from one to four.

Efficiency: The 1:1 configuration achieves the highest peak efficiency of 95.57%, which is expected since its single-coil geometry allows the most concentrated and symmetric flux coupling under perfectly aligned conditions. The 1:2 and 1:4 configurations both converge to a peak efficiency of approximately 94.79%, a marginal reduction of less than 1% from the 1:1 baseline. This near-identical efficiency between the 1:2 and 1:4 cases is noteworthy: it indicates that adding additional receiver coils beyond two does not further penalise efficiency, while continuing to deliver operational benefits in other dimensions.

Resonance Bandwidth: The 1:1 configuration exhibits a sharp, narrow resonance peak. While this maximises efficiency at exactly the design frequency, it makes the system sensitive to component tolerances and operating frequency deviations. The 1:2 and 1:4 configurations both show a broader resonance profile, which is more forgiving in practical hardware implementations where parasitic components and temperature-dependent inductance values can shift the effective resonant frequency slightly from its nominal design value.

Misalignment Tolerance: This is where the progression from 1:1 to 1:2 to 1:4 is most consequential. The 1:1 system's efficiency degrades sharply with even modest lateral offset between the transmitter and receiver coils. The 1:2 configuration improves upon this in the direction of receiver separation, but still remains vulnerable to misalignment perpendicular to the axis connecting the two receiver coils. The 1:4 configuration, with receivers distributed at all four corners of the drone's 50 cm × 50 cm frame, provides spatial coverage in all lateral directions, making it robust against landing offsets regardless of orientation.

Thermal Distribution: As the number of series-connected receiver coils increases, the induced current flowing through each individual coil decreases for the same total delivered power. This reduces I^2R losses per coil and prevents heat concentration at any single point in the receiver assembly, which is important for long-term reliability during extended charging cycles in outdoor agricultural conditions.

Practical Deployment: The 1:1 configuration, despite its highest peak efficiency, is fundamentally unsuitable for uncontrolled outdoor deployment because it demands near-perfect landing precision every time. The 1:2 configuration is an improvement and would be viable in semi-controlled settings, but its unidirectional misalignment tolerance leaves it exposed under real field conditions. The 1:4 configuration provides the most complete solution — omnidirectional misalignment tolerance, distributed thermal management, and series voltage addition — at the cost of only a 0.78% reduction in peak efficiency and a moderate increase in receiver-side assembly complexity.

The progression from 1:1 to 1:2 to 1:4 therefore tells a clear engineering story: each step sacrifices a negligible amount of peak efficiency in exchange for substantial and

practically meaningful gains in robustness, making the 1:4 configuration the appropriate choice for the wireless agricultural drone charging system developed in this project.

6.7 Summary

This chapter presented the coil design rationale, configuration options, and comparative simulation results for the wireless charging system. Circular coil geometry was selected for its axial symmetry, high quality factor, compatibility with drone geometry, and analytical tractability. Four general configuration classes — 1:1, 1:2, 1:4, and N:N — were reviewed, with the 1:1, 1:2, and 1:4 configurations taken forward for detailed simulation at 85 kHz.

Analytical calculations established 23 turns for both transmitting and receiving coils, which were refined through Ansys Maxwell 3D electromagnetic simulation and Ansys Twin Builder circuit co-simulation. All three configurations achieved peak efficiencies in the range of 94.79%–95.57% at the target frequency. However, the comparative analysis demonstrated that the 1:4 configuration uniquely combines high efficiency, omnidirectional misalignment tolerance, distributed thermal management, and a large effective charging footprint — all of which are essential for reliable autonomous operation in agricultural field conditions. The 1:2 configuration served as a valuable intermediate data point, confirming that the transition from single to multiple receivers does not significantly degrade efficiency while progressively improving practical robustness.

On this basis, the 1:4 coil configuration is adopted as the proposed design for this wireless drone charging system, with the subsequent chapters addressing the compensation circuit design and full system integration required to realise this configuration at the 2-kW target power rating.

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

This project successfully developed and validated a complete wireless power transfer system for the autonomous charging of agricultural drones. The work spanned the full signal chain from AC mains input through power factor correction, high-frequency inversion, resonant coil coupling, and multi-coil configuration analysis demonstrating a coherent and implementable system architecture.

The Totem-Pole PFC stage, particularly the fully active MOSFET variant, was shown through simulation to achieve a conversion efficiency of approximately 98.9% with a power factor approaching unity, confirming its suitability as the front-end converter for high-power wireless charging applications. The H-Bridge inverter operating at 85 kHz selected in alignment with SAE J2954 standards reliably produced the high-frequency AC excitation required to drive the resonant coil network. Across all three coil configurations examined, electromagnetic and circuit-level co-simulations in Ansys Maxwell 3D and Ansys Twin Builder confirmed peak power transfer efficiencies in the range of 94.79% to 95.57% at the design frequency. The 1:4 configuration was identified as the optimal design, offering omnidirectional misalignment tolerance and distributed thermal performance at the cost of only a marginal reduction in peak efficiency compared to the 1:1 baseline. Together, these results demonstrate that a fully autonomous, contact-free drone charging system delivering 2 kW at 48 V is practically achievable with the proposed architecture.

7.2 Future Scope

The system developed in this project holds significant commercial potential across multiple domains. The most immediate opportunity lies in precision agriculture, where charging pads at field stations could enable continuous drone relay missions without human intervention. The architecture is equally applicable to warehouse logistics drones, infrastructure inspection platforms monitoring power lines and pipelines, and urban air mobility systems all environments where autonomous, contact-free recharging eliminates critical operational bottlenecks.

On the technical roadmap, integration of adaptive frequency control and foreign object detection would enhance safety under variable field conditions. Scaling toward higher power ratings with bidirectional capability could enable vehicle-to-grid (V2G) functionality, allowing drone fleets to serve as distributed energy storage assets alongside their primary operational role.

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